Lesson 1.01 **Duration: 40 minutes**

What is Networking?

Learning Objectives

- Define computer networking and its purpose
- Identify real-world examples of networks
- Explain how devices communicate through a network

Engage (5-7 minutes)

Teacher Activity

Ask: ‘How do you search for something on Google? What makes one search better than another?’ Write tips on board. Discuss how effective searching saves time.

Student Activity

Share search experiences. Identify strategies: specific keywords, quotes, site: operator. Discuss frustration with bad search results.

Explore (10-12 minutes)

Teacher Activity

Give each pair a search challenge: ‘Find the population of Chitradurga district.’ Time them. Discuss which search terms worked best. Compare Google, Bing, DuckDuckGo.

Student Activity

Search for the answer. Try different search terms. Compare results from different engines. Note which was fastest.

Explain (10-12 minutes)

Teacher Activity

Define email components: address format username@domain, providers Gmail, Yahoo. Demonstrate composing an email: To, Subject, Body, Attach. Show 2FA for security.

Student Activity

Take notes on email format. Observe the demonstration. Understand why 2FA adds security.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Compose an email to your teacher requesting permission for a school trip. Include: subject line, greeting, body with reasons, closing.' If no email access, write the draft on paper.

Student Activity

Write the email or draft. Apply proper format: subject, greeting, body, signature. Review for clarity.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two search tips. 2. What is the format of an email address? 3. Why is 2FA important?'

Student Activity

Complete the exit ticket. Discuss answers with partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in What is Networking? - mathematical concepts
Science: Scientific applications of What is Networking? - scientific applications
Language: Writing about What is Networking? builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Network diagram
- connection equipment
- troubleshooting guide

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of What is Networking? concepts through practical exercises
Advanced: Apply What is Networking? concepts to solve complex problems independently
SEND: Practice What is Networking? skills with step-by-step teacher guidance

Formative Assessment

Method: Practical What is Networking? activity and oral questioning
CT Skill Assessed: Decomposition - breaking down What is Networking? into manageable steps

Dormitory Task (Paper-and-Pencil)

Draw a simple network diagram showing how two computers connect through the internet.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.02 **Duration: 40 minutes**

Types of Networks

Learning Objectives

- Differentiate between LAN, MAN, and WAN
- Give examples of each network type from daily life
- Understand the range and speed differences between network types

Engage (5-7 minutes)

Teacher Activity

Ask: ‘How do you search for something on Google? What makes one search better than another?’ Write tips on board. Discuss how effective searching saves time.

Student Activity

Share search experiences. Identify strategies: specific keywords, quotes, site: operator. Discuss frustration with bad search results.

Explore (10-12 minutes)

Teacher Activity

Give each pair a search challenge: ‘Find the population of Chitradurga district.’ Time them. Discuss which search terms worked best. Compare Google, Bing, DuckDuckGo.

Student Activity

Search for the answer. Try different search terms. Compare results from different engines. Note which was fastest.

Explain (10-12 minutes)

Teacher Activity

Define email components: address format username@domain, providers Gmail, Yahoo. Demonstrate composing an email: To, Subject, Body, Attach. Show 2FA for security.

Student Activity

Take notes on email format. Observe the demonstration. Understand why 2FA adds security.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Compose an email to your teacher requesting permission for a school trip. Include: subject line, greeting, body with reasons, closing.’ If no email access, write the draft on paper.

Student Activity

Write the email or draft. Apply proper format: subject, greeting, body, signature. Review for clarity.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. Name two search tips. 2. What is the format of an email address? 3. Why is 2FA important?’

Student Activity

Complete the exit ticket. Discuss answers with partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Types of Networks - mathematical concepts
Science: Scientific applications of Types of Networks - scientific applications
Language: Writing about Types of Networks builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Network diagram
- connection equipment
- troubleshooting guide

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Types of Networks concepts through practical exercises
Advanced: Apply Types of Networks concepts to solve complex problems independently
SEND: Practice Types of Networks skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Types of Networks activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Types of Networks into manageable steps

Dormitory Task (Paper-and-Pencil)

Draw a simple network diagram showing how two computers connect through the internet.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.03 **Duration: 40 minutes**

Network Devices

Learning Objectives

- Identify common network devices and their functions
- Explain the role of routers, switches, and modems
- Differentiate between wired and wireless devices

Engage (5-7 minutes)

Teacher Activity

Ask: 'How do you search for something on Google? What makes one search better than another?' Write tips on board. Discuss how effective searching saves time.

Student Activity

Share search experiences. Identify strategies: specific keywords, quotes, site: operator. Discuss frustration with bad search results.

Explore (10-12 minutes)

Teacher Activity

Give each pair a search challenge: 'Find the population of Chitradurga district.' Time them. Discuss which search terms worked best. Compare Google, Bing, DuckDuckGo.

Student Activity

Search for the answer. Try different search terms. Compare results from different engines. Note which was fastest.

Explain (10-12 minutes)

Teacher Activity

Define email components: address format username@domain, providers Gmail, Yahoo. Demonstrate composing an email: To, Subject, Body, Attach. Show 2FA for security.

Student Activity

Take notes on email format. Observe the demonstration. Understand why 2FA adds security.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Compose an email to your teacher requesting permission for a school trip. Include: subject line, greeting, body with reasons, closing.’ If no email access, write the draft on paper.

Student Activity

Write the email or draft. Apply proper format: subject, greeting, body, signature. Review for clarity.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. Name two search tips. 2. What is the format of an email address? 3. Why is 2FA important?’

Student Activity

Complete the exit ticket. Discuss answers with partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Network Devices - mathematical concepts
Science: Scientific applications of Network Devices - scientific applications
Language: Writing about Network Devices builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Network diagram
- connection equipment
- troubleshooting guide

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Network Devices concepts through practical exercises
Advanced: Apply Network Devices concepts to solve complex problems independently
SEND: Practice Network Devices skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Network Devices activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Network Devices into manageable steps

Dormitory Task (Paper-and-Pencil)

Draw a simple network diagram showing how two computers connect through the internet.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.04 **Duration: 40 minutes**

Internet Connection

Learning Objectives

- Identify different types of internet connections
- Compare broadband, mobile data, and Wi-Fi
- Understand the concept of bandwidth and speed

Engage (5-7 minutes)

Teacher Activity

Ask: ‘How do you search for something on Google? What makes one search better than another?’ Write tips on board. Discuss how effective searching saves time.

Student Activity

Share search experiences. Identify strategies: specific keywords, quotes, site: operator. Discuss frustration with bad search results.

Explore (10-12 minutes)

Teacher Activity

Give each pair a search challenge: ‘Find the population of Chitradurga district.’ Time them. Discuss which search terms worked best. Compare Google, Bing, DuckDuckGo.

Student Activity

Search for the answer. Try different search terms. Compare results from different engines. Note which was fastest.

Explain (10-12 minutes)

Teacher Activity

Define email components: address format username@domain, providers Gmail, Yahoo. Demonstrate composing an email: To, Subject, Body, Attach. Show 2FA for security.

Student Activity

Take notes on email format. Observe the demonstration. Understand why 2FA adds security.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Compose an email to your teacher requesting permission for a school trip. Include: subject line, greeting, body with reasons, closing.’ If no email access, write the draft on paper.

Student Activity

Write the email or draft. Apply proper format: subject, greeting, body, signature. Review for clarity.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. Name two search tips. 2. What is the format of an email address? 3. Why is 2FA important?’

Student Activity

Complete the exit ticket. Discuss answers with partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Internet Connection - mathematical concepts
Science: Scientific applications of Internet Connection - scientific applications
Language: Writing about Internet Connection builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Network diagram
- connection equipment
- troubleshooting guide

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Internet Connection concepts through practical exercises
Advanced: Apply Internet Connection concepts to solve complex problems independently
SEND: Practice Internet Connection skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Internet Connection activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Internet Connection into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Internet Connection. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.05 **Duration: 40 minutes**

Internet Protocols

Learning Objectives

- Define what an internet protocol is
- Explain the role of IP addresses and URLs
- Understand how HTTP and HTTPS work

Engage (5-7 minutes)

Teacher Activity

Ask: ‘How do you search for something on Google? What makes one search better than another?’ Write tips on board. Discuss how effective searching saves time.

Student Activity

Share search experiences. Identify strategies: specific keywords, quotes, site: operator. Discuss frustration with bad search results.

Explore (10-12 minutes)

Teacher Activity

Give each pair a search challenge: ‘Find the population of Chitradurga district.’ Time them. Discuss which search terms worked best. Compare Google, Bing, DuckDuckGo.

Student Activity

Search for the answer. Try different search terms. Compare results from different engines. Note which was fastest.

Explain (10-12 minutes)

Teacher Activity

Define email components: address format username@domain, providers Gmail, Yahoo. Demonstrate composing an email: To, Subject, Body, Attach. Show 2FA for security.

Student Activity

Take notes on email format. Observe the demonstration. Understand why 2FA adds security.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Compose an email to your teacher requesting permission for a school trip. Include: subject line, greeting, body with reasons, closing.' If no email access, write the draft on paper.

Student Activity

Write the email or draft. Apply proper format: subject, greeting, body, signature. Review for clarity.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two search tips. 2. What is the format of an email address? 3. Why is 2FA important?'

Student Activity

Complete the exit ticket. Discuss answers with partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Internet Protocols - mathematical concepts
Science: Scientific applications of Internet Protocols - scientific applications
Language: Writing about Internet Protocols builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Network diagram
- connection equipment
- troubleshooting guide

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Internet Protocols concepts through practical exercises
Advanced: Apply Internet Protocols concepts to solve complex problems independently
SEND: Practice Internet Protocols skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Internet Protocols activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Internet Protocols into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Internet Protocols. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.06 **Duration: 40 minutes**

Email Services

Learning Objectives

- Understand what email is and how it works
- Identify the parts of an email address
- Compose a basic email with subject, body, and recipient

Engage (5-7 minutes)

Teacher Activity

Ask: 'How do you search for something on Google? What makes one search better than another?' Write tips on board. Discuss how effective searching saves time.

Student Activity

Share search experiences. Identify strategies: specific keywords, quotes, site: operator. Discuss frustration with bad search results.

Explore (10-12 minutes)

Teacher Activity

Give each pair a search challenge: 'Find the population of Chitradurga district.' Time them. Discuss which search terms worked best. Compare Google, Bing, DuckDuckGo.

Student Activity

Search for the answer. Try different search terms. Compare results from different engines. Note which was fastest.

Explain (10-12 minutes)

Teacher Activity

Define email components: address format username@domain, providers Gmail, Yahoo. Demonstrate composing an email: To, Subject, Body, Attach. Show 2FA for security.

Student Activity

Take notes on email format. Observe the demonstration. Understand why 2FA adds security.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Compose an email to your teacher requesting permission for a school trip. Include: subject line, greeting, body with reasons, closing.’ If no email access, write the draft on paper.

Student Activity

Write the email or draft. Apply proper format: subject, greeting, body, signature. Review for clarity.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. Name two search tips. 2. What is the format of an email address? 3. Why is 2FA important?’

Student Activity

Complete the exit ticket. Discuss answers with partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Email Services - mathematical concepts
Science: Scientific applications of Email Services - scientific applications
Language: Writing about Email Services builds technical vocabulary and communication skills

Digital Citizen Reminder: Do not open emails from strangers - they may contain viruses.

Resources Needed:

- Research materials
- example AI applications
- discussion guide

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Email Services concepts through practical exercises
Advanced: Apply Email Services concepts to solve complex problems independently
SEND: Practice Email Services skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Email Services activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Email Services into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a paragraph explaining how Email Services is used in real life. Give two examples.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.07 **Duration: 40 minutes**

Attachments

Learning Objectives

- Attach files to an email correctly
- Understand file size limits and common formats
- Open and save email attachments safely

Engage (5-7 minutes)

Teacher Activity

Ask: ‘How do you search for something on Google? What makes one search better than another?’ Write tips on board. Discuss how effective searching saves time.

Student Activity

Share search experiences. Identify strategies: specific keywords, quotes, site: operator. Discuss frustration with bad search results.

Explore (10-12 minutes)

Teacher Activity

Give each pair a search challenge: ‘Find the population of Chitradurga district.’ Time them. Discuss which search terms worked best. Compare Google, Bing, DuckDuckGo.

Student Activity

Search for the answer. Try different search terms. Compare results from different engines. Note which was fastest.

Explain (10-12 minutes)

Teacher Activity

Define email components: address format username@domain, providers Gmail, Yahoo. Demonstrate composing an email: To, Subject, Body, Attach. Show 2FA for security.

Student Activity

Take notes on email format. Observe the demonstration. Understand why 2FA adds security.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Compose an email to your teacher requesting permission for a school trip. Include: subject line, greeting, body with reasons, closing.' If no email access, write the draft on paper.

Student Activity

Write the email or draft. Apply proper format: subject, greeting, body, signature. Review for clarity.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two search tips. 2. What is the format of an email address? 3. Why is 2FA important?'

Student Activity

Complete the exit ticket. Discuss answers with partner.

Cross-Curricular Connections

Mathematics: File sizes affect download time - understanding data transfer rates
Science: File formats serve different purposes - technology standards
Language: Writing clear attachment descriptions builds precise language

Digital Citizen Reminder: Only open attachments from people you trust - they may contain viruses.

Resources Needed:

- Email with attachments
- file type guide
- practice files

Differentiation

Approaching: Read an email and identify the attachment
On Level: Attach a file to an email and send it
Advanced: Attach multiple files of different formats with descriptions
SEND: Attach a file to an email with teacher guidance

Formative Assessment

Method: Attachment task
CT Skill Assessed: Algorithm Design - following steps to attach and send files

Dormitory Task (Paper-and-Pencil)

Write instructions for attaching a file to an email. List at least 5 steps.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.08	Duration: 40 minutes

Internet Safety

Learning Objectives

- Identify common online threats and scams
- Practice safe browsing habits
- Understand the importance of strong passwords

Engage (5-7 minutes)

Teacher Activity

Ask: ‘How do you search for something on Google? What makes one search better than another?’ Write tips on board. Discuss how effective searching saves time.

Student Activity

Share search experiences. Identify strategies: specific keywords, quotes, site: operator. Discuss frustration with bad search results.

Explore (10-12 minutes)

Teacher Activity

Give each pair a search challenge: ‘Find the population of Chitradurga district.’ Time them. Discuss which search terms worked best. Compare Google, Bing, DuckDuckGo.

Student Activity

Search for the answer. Try different search terms. Compare results from different engines. Note which was fastest.

Explain (10-12 minutes)

Teacher Activity

Define email components: address format username@domain, providers Gmail, Yahoo. Demonstrate composing an email: To, Subject, Body, Attach. Show 2FA for security.

Student Activity

Take notes on email format. Observe the demonstration. Understand why 2FA adds security.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Compose an email to your teacher requesting permission for a school trip. Include: subject line, greeting, body with reasons, closing.' If no email access, write the draft on paper.

Student Activity

Write the email or draft. Apply proper format: subject, greeting, body, signature. Review for clarity.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two search tips. 2. What is the format of an email address? 3. Why is 2FA important?'

Student Activity

Complete the exit ticket. Discuss answers with partner.

Cross-Curricular Connections

Mathematics: Risk assessment in math - probability and consequences
Science: Understanding online threats relates to biology (viruses as metaphor)
Language: Writing safety rules builds persuasive and instructional writing

Digital Citizen Reminder: Never share your password with anyone except your parents or teacher. Use strong passwords with letters, numbers, and symbols.

Resources Needed:
- Safety rules poster
- scenario cards
- pledge template

Differentiation

Approaching: Match online safety rules to their correct category
On Level: Identify safe and unsafe behaviors in five online scenarios
Advanced: Create a personal internet safety plan with at least 8 rules
SEND: Discuss safety rules with teacher guidance using picture prompts

Formative Assessment

Method: Safety pledge and scenario quiz
CT Skill Assessed: Evaluation - assessing online situations for safety risks

Dormitory Task (Paper-and-Pencil)

Write ten internet safety rules. For each rule, explain what could happen if you do not follow it.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.09 **Duration: 40 minutes**

Social Media

Learning Objectives

- Identify popular social media platforms
- Understand the benefits and risks of social media
- Practice responsible social media behavior

Engage (5-7 minutes)

Teacher Activity

Ask: 'How do you search for something on Google? What makes one search better than another?' Write tips on board. Discuss how effective searching saves time.

Student Activity

Share search experiences. Identify strategies: specific keywords, quotes, site: operator. Discuss frustration with bad search results.

Explore (10-12 minutes)

Teacher Activity

Give each pair a search challenge: 'Find the population of Chitradurga district.' Time them. Discuss which search terms worked best. Compare Google, Bing, DuckDuckGo.

Student Activity

Search for the answer. Try different search terms. Compare results from different engines. Note which was fastest.

Explain (10-12 minutes)

Teacher Activity

Define email components: address format username@domain, providers Gmail, Yahoo. Demonstrate composing an email: To, Subject, Body, Attach. Show 2FA for security.

Student Activity

Take notes on email format. Observe the demonstration. Understand why 2FA adds security.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Compose an email to your teacher requesting permission for a school trip. Include: subject line, greeting, body with reasons, closing.’ If no email access, write the draft on paper.

Student Activity

Write the email or draft. Apply proper format: subject, greeting, body, signature. Review for clarity.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. Name two search tips. 2. What is the format of an email address? 3. Why is 2FA important?’

Student Activity

Complete the exit ticket. Discuss answers with partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Social Media - mathematical concepts
Science: Scientific applications of Social Media - scientific applications
Language: Writing about Social Media builds technical vocabulary and communication skills

Digital Citizen Reminder: Be kind and respectful on social media.

Resources Needed:
- Chapter 1 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Social Media concepts through practical exercises
Advanced: Apply Social Media concepts to solve complex problems independently
SEND: Practice Social Media skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Social Media activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Social Media into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Social Media. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.10 | **Duration: 40 minutes**

Download Management

Learning Objectives

- Download files safely from the internet
- Understand file types and where downloads are saved
- Organize and manage downloaded files

Engage (5-7 minutes)

Teacher Activity

Ask: 'How do you search for something on Google? What makes one search better than another?' Write tips on board. Discuss how effective searching saves time.

Student Activity

Share search experiences. Identify strategies: specific keywords, quotes, site: operator. Discuss frustration with bad search results.

Explore (10-12 minutes)

Teacher Activity

Give each pair a search challenge: 'Find the population of Chitradurga district.' Time them. Discuss which search terms worked best. Compare Google, Bing, DuckDuckGo.

Student Activity

Search for the answer. Try different search terms. Compare results from different engines. Note which was fastest.

Explain (10-12 minutes)

Teacher Activity

Define email components: address format username@domain, providers Gmail, Yahoo. Demonstrate composing an email: To, Subject, Body, Attach. Show 2FA for security.

Student Activity

Take notes on email format. Observe the demonstration. Understand why 2FA adds security.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Compose an email to your teacher requesting permission for a school trip. Include: subject line, greeting, body with reasons, closing.’ If no email access, write the draft on paper.

Student Activity

Write the email or draft. Apply proper format: subject, greeting, body, signature. Review for clarity.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. Name two search tips. 2. What is the format of an email address? 3. Why is 2FA important?’

Student Activity

Complete the exit ticket. Discuss answers with partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Download Management - mathematical concepts
Science: Scientific applications of Download Management - scientific applications
Language: Writing about Download Management builds technical vocabulary and communication skills

Digital Citizen Reminder: Only download files from trusted websites.

Resources Needed:
- Chapter 1 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Download Management concepts through practical exercises
Advanced: Apply Download Management concepts to solve complex problems independently
SEND: Practice Download Management skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Download Management activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Download Management into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Download Management. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.11 **Duration: 40 minutes**

Network Troubleshooting

Learning Objectives

- Identify common network problems
- Use basic troubleshooting steps
- Understand when to seek technical help

Engage (5-7 minutes)

Teacher Activity

Ask: ‘How do you search for something on Google? What makes one search better than another?’ Write tips on board. Discuss how effective searching saves time.

Student Activity

Share search experiences. Identify strategies: specific keywords, quotes, site: operator. Discuss frustration with bad search results.

Explore (10-12 minutes)

Teacher Activity

Give each pair a search challenge: ‘Find the population of Chitradurga district.’ Time them. Discuss which search terms worked best. Compare Google, Bing, DuckDuckGo.

Student Activity

Search for the answer. Try different search terms. Compare results from different engines. Note which was fastest.

Explain (10-12 minutes)

Teacher Activity

Define email components: address format username@domain, providers Gmail, Yahoo. Demonstrate composing an email: To, Subject, Body, Attach. Show 2FA for security.

Student Activity

Take notes on email format. Observe the demonstration. Understand why 2FA adds security.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Compose an email to your teacher requesting permission for a school trip. Include: subject line, greeting, body with reasons, closing.’ If no email access, write the draft on paper.

Student Activity

Write the email or draft. Apply proper format: subject, greeting, body, signature. Review for clarity.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. Name two search tips. 2. What is the format of an email address? 3. Why is 2FA important?’

Student Activity

Complete the exit ticket. Discuss answers with partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Network Troubleshooting - mathematical concepts
Science: Scientific applications of Network Troubleshooting - scientific applications
Language: Writing about Network Troubleshooting builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Network diagram
- connection equipment
- troubleshooting guide

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Network Troubleshooting concepts through practical exercises
Advanced: Apply Network Troubleshooting concepts to solve complex problems independently
SEND: Practice Network Troubleshooting skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Network Troubleshooting activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Network Troubleshooting into manageable steps

Dormitory Task (Paper-and-Pencil)

Draw a simple network diagram showing how two computers connect through the internet.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.12 **Duration: 40 minutes**

Chapter 1 Review

Learning Objectives

- Review all Chapter 1 networking concepts
- Connect networking topics to real-world applications
- Identify areas needing further practice

Engage (5-7 minutes)

Teacher Activity

Ask: ‘How do you search for something on Google? What makes one search better than another?’ Write tips on board. Discuss how effective searching saves time.

Student Activity

Share search experiences. Identify strategies: specific keywords, quotes, site: operator. Discuss frustration with bad search results.

Explore (10-12 minutes)

Teacher Activity

Give each pair a search challenge: ‘Find the population of Chitradurga district.’ Time them. Discuss which search terms worked best. Compare Google, Bing, DuckDuckGo.

Student Activity

Search for the answer. Try different search terms. Compare results from different engines. Note which was fastest.

Explain (10-12 minutes)

Teacher Activity

Define email components: address format username@domain, providers Gmail, Yahoo. Demonstrate composing an email: To, Subject, Body, Attach. Show 2FA for security.

Student Activity

Take notes on email format. Observe the demonstration. Understand why 2FA adds security.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Compose an email to your teacher requesting permission for a school trip. Include: subject line, greeting, body with reasons, closing.' If no email access, write the draft on paper.

Student Activity

Write the email or draft. Apply proper format: subject, greeting, body, signature. Review for clarity.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two search tips. 2. What is the format of an email address? 3. Why is 2FA important?'

Student Activity

Complete the exit ticket. Discuss answers with partner.

Cross-Curricular Connections

Mathematics: Review exercises reinforce mathematical problem-solving
Science: Chapter review connects all internet and computing concepts
Language: Writing summaries builds synthesis and review skills

Digital Citizen Reminder: Apply internet safety rules you learned in every online activity.

Resources Needed:
- Review worksheet
- answer key
- chapter summary

Differentiation

Approaching: Complete fill-in-the-blank review questions
On Level: Answer short answer questions about chapter concepts
Advanced: Create a concept map connecting all chapter topics
SEND: Review key terms with teacher guidance using flashcards

Formative Assessment

Method: Chapter review quiz
CT Skill Assessed: Evaluation - self-assessing understanding of chapter concepts

Dormitory Task (Paper-and-Pencil)

Write a summary of everything you learned in this chapter. Include at least 8 key points.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.01 **Duration: 40 minutes**

Introduction to GIMP

Learning Objectives

- Open and navigate the GIMP interface
- Identify key toolbox elements and menus
- Create and save a new image file

Engage (5-7 minutes)

Teacher Activity

Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What changes do you see? How do you think this was done?' Introduce GIMP as a free image editor.

Student Activity

Compare both images. Identify changes: crop, rotation, brightness. Express interest in learning the tools.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: crop tool, rotate, brightness/contrast, resize. Students follow along with a sample image.

Student Activity

Follow along. Apply each tool to the sample image. Experiment with different settings.

Explain (10-12 minutes)

Teacher Activity

Define layers: transparent sheets stacked to create the final image. Each layer holds separate elements. Show the Layers panel. Demonstrate: create layer, draw on it, toggle visibility, adjust opacity.

Student Activity

Take notes. Understand that layers allow editing one part without affecting others. Practice creating and managing layers.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple composition with background on one layer and text on another. Adjust text opacity to 50%.' If no computer, sketch the layer concept on paper.

Student Activity

Work on the task. Create layers, add content, adjust properties. Get peer feedback.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is a layer? 2. Name two image editing operations. 3. Why are layers useful?'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Introduction to GIMP - geometric transformations
Science: Scientific applications of Introduction to GIMP - digital imaging
Language: Writing about Introduction to GIMP builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:
- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support
On Level: Demonstrate understanding of Introduction to GIMP concepts through practical exercises
Advanced: Apply Introduction to GIMP concepts to solve complex problems independently
SEND: Practice Introduction to GIMP skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Introduction to GIMP activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Introduction to GIMP into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Introduction to GIMP. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.02 **Duration: 40 minutes**

Selection Tools

Learning Objectives

- Use rectangle, ellipse, and free select tools
- Understand when to use each selection type
- Practice selecting and modifying portions of an image

Engage (5-7 minutes)

Teacher Activity

Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What changes do you see? How do you think this was done?' Introduce GIMP as a free image editor.

Student Activity

Compare both images. Identify changes: crop, rotation, brightness. Express interest in learning the tools.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: crop tool, rotate, brightness/contrast, resize. Students follow along with a sample image.

Student Activity

Follow along. Apply each tool to the sample image. Experiment with different settings.

Explain (10-12 minutes)

Teacher Activity

Define layers: transparent sheets stacked to create the final image. Each layer holds separate elements. Show the Layers panel. Demonstrate: create layer, draw on it, toggle visibility, adjust opacity.

Student Activity

Take notes. Understand that layers allow editing one part without affecting others. Practice creating and managing layers.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple composition with background on one layer and text on another. Adjust text opacity to 50%.' If no computer, sketch the layer concept on paper.

Student Activity

Work on the task. Create layers, add content, adjust properties. Get peer feedback.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is a layer? 2. Name two image editing operations. 3. Why are layers useful?'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Selection Tools - geometric transformations
Science: Scientific applications of Selection Tools - digital imaging
Language: Writing about Selection Tools builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:
- Advanced Graphics reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support
On Level: Demonstrate understanding of Selection Tools concepts through practical exercises
Advanced: Apply Selection Tools concepts to solve complex problems independently
SEND: Practice Selection Tools skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Selection Tools activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Selection Tools into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Selection Tools. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.03 **Duration: 40 minutes**

Layers Concept

Learning Objectives

- Understand what layers are and why they matter
- Create, rename, and reorder layers
- Use layer visibility and opacity controls

Engage (5-7 minutes)

Teacher Activity

Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What changes do you see? How do you think this was done?' Introduce GIMP as a free image editor.

Student Activity

Compare both images. Identify changes: crop, rotation, brightness. Express interest in learning the tools.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: crop tool, rotate, brightness/contrast, resize. Students follow along with a sample image.

Student Activity

Follow along. Apply each tool to the sample image. Experiment with different settings.

Explain (10-12 minutes)

Teacher Activity

Define layers: transparent sheets stacked to create the final image. Each layer holds separate elements. Show the Layers panel. Demonstrate: create layer, draw on it, toggle visibility, adjust opacity.

Student Activity

Take notes. Understand that layers allow editing one part without affecting others. Practice creating and managing layers.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple composition with background on one layer and text on another. Adjust text opacity to 50%.' If no computer, sketch the layer concept on paper.

Student Activity

Work on the task. Create layers, add content, adjust properties. Get peer feedback.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is a layer? 2. Name two image editing operations. 3. Why are layers useful?'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: Layers stack like fractions - understanding ordering and precedence
Science: How transparent layers work - optics and light
Language: Explaining layers requires clear organizational writing

Digital Citizen Reminder: Understanding layers helps organize work - a skill useful beyond computing.

Resources Needed:
- Computer with layer-capable editor
- layer examples
- practice files

Differentiation

Approaching: Create two layers and draw different elements on each
On Level: Use layers to create a composite image with foreground and background
Advanced: Manage complex multi-layer compositions with blending modes
SEND: Create and manage layers with teacher guidance

Formative Assessment

Method: Layer management demonstration
CT Skill Assessed: Abstraction - understanding that layers separate elements for independent editing

Dormitory Task (Paper-and-Pencil)

Draw three separate elements (sun, house, tree) on separate pieces of paper. Stack them to create one picture.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.04 **Duration: 40 minutes**

Layer Effects

Learning Objectives

- Apply drop shadows and glows to layers
- Use layer masks for non-destructive editing
- Combine multiple layer effects for creative results

Engage (5-7 minutes)

Teacher Activity

Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What changes do you see? How do you think this was done?' Introduce GIMP as a free image editor.

Student Activity

Compare both images. Identify changes: crop, rotation, brightness. Express interest in learning the tools.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: crop tool, rotate, brightness/contrast, resize. Students follow along with a sample image.

Student Activity

Follow along. Apply each tool to the sample image. Experiment with different settings.

Explain (10-12 minutes)

Teacher Activity

Define layers: transparent sheets stacked to create the final image. Each layer holds separate elements. Show the Layers panel. Demonstrate: create layer, draw on it, toggle visibility, adjust opacity.

Student Activity

Take notes. Understand that layers allow editing one part without affecting others. Practice creating and managing layers.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple composition with background on one layer and text on another. Adjust text opacity to 50%.' If no computer, sketch the layer concept on paper.

Student Activity

Work on the task. Create layers, add content, adjust properties. Get peer feedback.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is a layer? 2. Name two image editing operations. 3. Why are layers useful?'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Layer Effects - geometric transformations
Science: Scientific applications of Layer Effects - digital imaging
Language: Writing about Layer Effects builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:
- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support
On Level: Demonstrate understanding of Layer Effects concepts through practical exercises
Advanced: Apply Layer Effects concepts to solve complex problems independently
SEND: Practice Layer Effects skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Layer Effects activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Layer Effects into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Layer Effects. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.05 **Duration: 40 minutes**

Drawing Tools

Learning Objectives

- Use pencil, paintbrush, and airbrush tools
- Adjust brush size, shape, and opacity
- Create freehand drawings and simple illustrations

Engage (5-7 minutes)

Teacher Activity

Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What changes do you see? How do you think this was done?' Introduce GIMP as a free image editor.

Student Activity

Compare both images. Identify changes: crop, rotation, brightness. Express interest in learning the tools.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: crop tool, rotate, brightness/contrast, resize. Students follow along with a sample image.

Student Activity

Follow along. Apply each tool to the sample image. Experiment with different settings.

Explain (10-12 minutes)

Teacher Activity

Define layers: transparent sheets stacked to create the final image. Each layer holds separate elements. Show the Layers panel. Demonstrate: create layer, draw on it, toggle visibility, adjust opacity.

Student Activity

Take notes. Understand that layers allow editing one part without affecting others. Practice creating and managing layers.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple composition with background on one layer and text on another. Adjust text opacity to 50%.' If no computer, sketch the layer concept on paper.

Student Activity

Work on the task. Create layers, add content, adjust properties. Get peer feedback.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is a layer? 2. Name two image editing operations. 3. Why are layers useful?'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: Geometric shapes in art - identifying mathematical shapes in drawings
Science: Understanding brush strokes relates to fluid dynamics
Language: Describing drawing techniques builds technical vocabulary

Digital Citizen Reminder: Use drawing tools responsibly - do not draw on screens with markers or sharp objects.

Resources Needed:
- Computer with Paint
- tool practice sheets
- example artwork

Differentiation

Approaching: Use the line and rectangle tools to draw a building
On Level: Create a picture using at least 6 different drawing tools
Advanced: Combine advanced tools to create a detailed scene
SEND: Practice drawing with each tool and describe the results

Formative Assessment

Method: Drawing portfolio and tool checklist
CT Skill Assessed: Pattern Recognition - using repeated elements in designs

Dormitory Task (Paper-and-Pencil)

Draw a picture using only straight lines and rectangles. Count and write how many of each you used.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.06 **Duration: 40 minutes**

Color Adjustments

Learning Objectives

- Adjust brightness, contrast, and color balance
- Use hue-saturation to modify colors
- Correct common photo problems like underexposure

Engage (5-7 minutes)

Teacher Activity

Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What changes do you see? How do you think this was done?' Introduce GIMP as a free image editor.

Student Activity

Compare both images. Identify changes: crop, rotation, brightness. Express interest in learning the tools.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: crop tool, rotate, brightness/contrast, resize. Students follow along with a sample image.

Student Activity

Follow along. Apply each tool to the sample image. Experiment with different settings.

Explain (10-12 minutes)

Teacher Activity

Define layers: transparent sheets stacked to create the final image. Each layer holds separate elements. Show the Layers panel. Demonstrate: create layer, draw on it, toggle visibility, adjust opacity.

Student Activity

Take notes. Understand that layers allow editing one part without affecting others. Practice creating and managing layers.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple composition with background on one layer and text on another. Adjust text opacity to 50%.' If no computer, sketch the layer concept on paper.

Student Activity

Work on the task. Create layers, add content, adjust properties. Get peer feedback.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is a layer? 2. Name two image editing operations. 3. Why are layers useful?'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Color Adjustments - geometric transformations
Science: Scientific applications of Color Adjustments - digital imaging
Language: Writing about Color Adjustments builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:
- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support
On Level: Demonstrate understanding of Color Adjustments concepts through practical exercises
Advanced: Apply Color Adjustments concepts to solve complex problems independently
SEND: Practice Color Adjustments skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Color Adjustments activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Color Adjustments into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Color Adjustments. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.07 **Duration: 40 minutes**

Filters Introduction

Learning Objectives

- Understand what filters do to images
- Apply blur, sharpen, and noise filters
- Use the preview feature before applying filters

Engage (5-7 minutes)

Teacher Activity

Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What changes do you see? How do you think this was done?' Introduce GIMP as a free image editor.

Student Activity

Compare both images. Identify changes: crop, rotation, brightness. Express interest in learning the tools.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: crop tool, rotate, brightness/contrast, resize. Students follow along with a sample image.

Student Activity

Follow along. Apply each tool to the sample image. Experiment with different settings.

Explain (10-12 minutes)

Teacher Activity

Define layers: transparent sheets stacked to create the final image. Each layer holds separate elements. Show the Layers panel. Demonstrate: create layer, draw on it, toggle visibility, adjust opacity.

Student Activity

Take notes. Understand that layers allow editing one part without affecting others. Practice creating and managing layers.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple composition with background on one layer and text on another. Adjust text opacity to 50%.' If no computer, sketch the layer concept on paper.

Student Activity

Work on the task. Create layers, add content, adjust properties. Get peer feedback.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is a layer? 2. Name two image editing operations. 3. Why are layers useful?'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Filters Introduction - geometric transformations
Science: Scientific applications of Filters Introduction - digital imaging
Language: Writing about Filters Introduction builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:
- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support
On Level: Demonstrate understanding of Filters Introduction concepts through practical exercises
Advanced: Apply Filters Introduction concepts to solve complex problems independently
SEND: Practice Filters Introduction skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Filters Introduction activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Filters Introduction into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Filters Introduction. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.08 | **Duration: 40 minutes**

Artistic Filters

Learning Objectives

- Apply oil painting and watercolor effects
- Use the cartoon filter for stylized images
- Combine multiple artistic filters creatively

Engage (5-7 minutes)

Teacher Activity

Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What changes do you see? How do you think this was done?' Introduce GIMP as a free image editor.

Student Activity

Compare both images. Identify changes: crop, rotation, brightness. Express interest in learning the tools.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: crop tool, rotate, brightness/contrast, resize. Students follow along with a sample image.

Student Activity

Follow along. Apply each tool to the sample image. Experiment with different settings.

Explain (10-12 minutes)

Teacher Activity

Define layers: transparent sheets stacked to create the final image. Each layer holds separate elements. Show the Layers panel. Demonstrate: create layer, draw on it, toggle visibility, adjust opacity.

Student Activity

Take notes. Understand that layers allow editing one part without affecting others. Practice creating and managing layers.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple composition with background on one layer and text on another. Adjust text opacity to 50%.' If no computer, sketch the layer concept on paper.

Student Activity

Work on the task. Create layers, add content, adjust properties. Get peer feedback.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is a layer? 2. Name two image editing operations. 3. Why are layers useful?'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Artistic Filters - geometric transformations
Science: Scientific applications of Artistic Filters - digital imaging
Language: Writing about Artistic Filters builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:

- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support
On Level: Demonstrate understanding of Artistic Filters concepts through practical exercises
Advanced: Apply Artistic Filters concepts to solve complex problems independently
SEND: Practice Artistic Filters skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Artistic Filters activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Artistic Filters into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Artistic Filters. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.09 **Duration: 40 minutes**

Text and Typography

Learning Objectives

- Add and format text on images
- Change font, size, color, and alignment
- Apply text effects like shadows and outlines

Engage (5-7 minutes)

Teacher Activity

Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What changes do you see? How do you think this was done?' Introduce GIMP as a free image editor.

Student Activity

Compare both images. Identify changes: crop, rotation, brightness. Express interest in learning the tools.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: crop tool, rotate, brightness/contrast, resize. Students follow along with a sample image.

Student Activity

Follow along. Apply each tool to the sample image. Experiment with different settings.

Explain (10-12 minutes)

Teacher Activity

Define layers: transparent sheets stacked to create the final image. Each layer holds separate elements. Show the Layers panel. Demonstrate: create layer, draw on it, toggle visibility, adjust opacity.

Student Activity

Take notes. Understand that layers allow editing one part without affecting others. Practice creating and managing layers.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple composition with background on one layer and text on another. Adjust text opacity to 50%.' If no computer, sketch the layer concept on paper.

Student Activity

Work on the task. Create layers, add content, adjust properties. Get peer feedback.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is a layer? 2. Name two image editing operations. 3. Why are layers useful?'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Text and Typography - geometric transformations
Science: Scientific applications of Text and Typography - digital imaging
Language: Writing about Text and Typography builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:
- Advanced Graphics reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support
On Level: Demonstrate understanding of Text and Typography concepts through practical exercises
Advanced: Apply Text and Typography concepts to solve complex problems independently
SEND: Practice Text and Typography skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Text and Typography activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Text and Typography into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Text and Typography. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.10 **Duration: 40 minutes**

Image Composition

Learning Objectives

- Combine multiple images into one composition
- Use cropping and resizing for proper layout
- Apply the rule of thirds for balanced design

Engage (5-7 minutes)

Teacher Activity

Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What changes do you see? How do you think this was done?' Introduce GIMP as a free image editor.

Student Activity

Compare both images. Identify changes: crop, rotation, brightness. Express interest in learning the tools.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: crop tool, rotate, brightness/contrast, resize. Students follow along with a sample image.

Student Activity

Follow along. Apply each tool to the sample image. Experiment with different settings.

Explain (10-12 minutes)

Teacher Activity

Define layers: transparent sheets stacked to create the final image. Each layer holds separate elements. Show the Layers panel. Demonstrate: create layer, draw on it, toggle visibility, adjust opacity.

Student Activity

Take notes. Understand that layers allow editing one part without affecting others. Practice creating and managing layers.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple composition with background on one layer and text on another. Adjust text opacity to 50%.' If no computer, sketch the layer concept on paper.

Student Activity

Work on the task. Create layers, add content, adjust properties. Get peer feedback.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is a layer? 2. Name two image editing operations. 3. Why are layers useful?'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Image Composition - geometric transformations
Science: Scientific applications of Image Composition - digital imaging
Language: Writing about Image Composition builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:

- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support
On Level: Demonstrate understanding of Image Composition concepts through practical exercises
Advanced: Apply Image Composition concepts to solve complex problems independently
SEND: Practice Image Composition skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Image Composition activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Image Composition into manageable steps

Dormitory Task (Paper-and-Pencil)

Draw a simple design using only basic shapes. Label the tools you would use to create it digitally.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.11 **Duration: 40 minutes**

Project - Photo Editing

Learning Objectives

- Apply all learned GIMP skills to edit a photo
- Practice selection, layers, filters, and text together
- Present a completed photo editing project

Engage (5-7 minutes)

Teacher Activity

Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What changes do you see? How do you think this was done?' Introduce GIMP as a free image editor.

Student Activity

Compare both images. Identify changes: crop, rotation, brightness. Express interest in learning the tools.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: crop tool, rotate, brightness/contrast, resize. Students follow along with a sample image.

Student Activity

Follow along. Apply each tool to the sample image. Experiment with different settings.

Explain (10-12 minutes)

Teacher Activity

Define layers: transparent sheets stacked to create the final image. Each layer holds separate elements. Show the Layers panel. Demonstrate: create layer, draw on it, toggle visibility, adjust opacity.

Student Activity

Take notes. Understand that layers allow editing one part without affecting others. Practice creating and managing layers.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple composition with background on one layer and text on another. Adjust text opacity to 50%.' If no computer, sketch the layer concept on paper.

Student Activity

Work on the task. Create layers, add content, adjust properties. Get peer feedback.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is a layer? 2. Name two image editing operations. 3. Why are layers useful?'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Photo Editing - geometric transformations
Science: Scientific applications of Project - Photo Editing - digital imaging
Language: Writing about Project - Photo Editing builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:

- Advanced Graphics reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support
On Level: Demonstrate understanding of Project - Photo Editing concepts through practical exercises
Advanced: Apply Project - Photo Editing concepts to solve complex problems independently
SEND: Practice Project - Photo Editing skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment
CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Photo Editing project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Photo Editing project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.12 **Duration: 40 minutes**

Chapter 2 Review

Learning Objectives

- Review all Chapter 2 GIMP concepts
- Connect graphics skills to real-world applications
- Identify areas needing further practice

Engage (5-7 minutes)

Teacher Activity

Show a photo and its cropped, rotated, and color-adjusted version. Ask: 'What changes do you see? How do you think this was done?' Introduce GIMP as a free image editor.

Student Activity

Compare both images. Identify changes: crop, rotation, brightness. Express interest in learning the tools.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: crop tool, rotate, brightness/contrast, resize. Students follow along with a sample image.

Student Activity

Follow along. Apply each tool to the sample image. Experiment with different settings.

Explain (10-12 minutes)

Teacher Activity

Define layers: transparent sheets stacked to create the final image. Each layer holds separate elements. Show the Layers panel. Demonstrate: create layer, draw on it, toggle visibility, adjust opacity.

Student Activity

Take notes. Understand that layers allow editing one part without affecting others. Practice creating and managing layers.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple composition with background on one layer and text on another. Adjust text opacity to 50%.' If no computer, sketch the layer concept on paper.

Student Activity

Work on the task. Create layers, add content, adjust properties. Get peer feedback.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is a layer? 2. Name two image editing operations. 3. Why are layers useful?'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: Review exercises consolidate programming concepts
Science: Chapter summaries connect related topics
Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 2 Review skills responsibly and ethically.

Resources Needed:
- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 2 Review support
On Level: Demonstrate Chapter 2 Review through practical exercises
Advanced: Apply Chapter 2 Review to solve complex problems independently
SEND: Practice Chapter 2 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 2 Review activity
CT Skill Assessed: Decomposition - breaking down Chapter 2 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 2 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.01 **Duration: 40 minutes**

What is Multimedia?

Learning Objectives

- Define multimedia and its components
- Identify text, audio, image, video, and animation as media types
- Explain how multimedia is used in daily life

Engage (5-7 minutes)

Teacher Activity

Show a short video clip 30 seconds. Ask: 'How do videos communicate messages better than text?' Discuss: visuals, audio, movement, emotion.

Student Activity

Watch the clip. Identify what makes it effective: music, transitions, text overlays. Share favorite video types.

Explore (10-12 minutes)

Teacher Activity

Show OpenShot interface or describe it. Demonstrate: import media, add to timeline, split clip, add transition. Students observe.

Student Activity

Observe the demonstration. Identify timeline, preview, and tools. Note how transitions improve flow.

Explain (10-12 minutes)

Teacher Activity

Define video communication: sharing information through moving images and sound. File formats: MP4, AVI, MOV. Video creation process: idea ⟶ script ⟶ shoot ⟶ edit ⟶ export.

Student Activity

Take notes. Understand the video creation workflow. Compare different file formats.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Plan a 1-minute video about your school. Write the script, list shots needed, and plan transitions.' If no computer, create a storyboard on paper.

Student Activity

Work in groups. Write script, draw storyboard, plan transitions. Present plan to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What are the steps to create a video? 2. Name two video file formats. 3. Why are transitions important?'

Student Activity

Complete the exit ticket. Review the creation process.

Cross-Curricular Connections

Mathematics: Mathematical concepts in What is Multimedia? - network topology
Science: Scientific applications of What is Multimedia? - data transmission
Language: Writing about What is Multimedia? builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of What is Multimedia? concepts through practical exercises
Advanced: Apply What is Multimedia? concepts to solve complex problems independently
SEND: Practice What is Multimedia? skills with step-by-step teacher guidance

Formative Assessment

Method: Practical What is Multimedia? activity and oral questioning
CT Skill Assessed: Decomposition - breaking down What is Multimedia? into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about What is Multimedia?. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.02 **Duration: 40 minutes**

Audio Editing with Audacity

Learning Objectives

- Open Audacity and identify its interface components
- Record, play, and save a short audio clip
- Apply basic trimming and cutting operations on audio

Engage (5-7 minutes)

Teacher Activity

Show a short video clip 30 seconds. Ask: 'How do videos communicate messages better than text?' Discuss: visuals, audio, movement, emotion.

Student Activity

Watch the clip. Identify what makes it effective: music, transitions, text overlays. Share favorite video types.

Explore (10-12 minutes)

Teacher Activity

Show OpenShot interface or describe it. Demonstrate: import media, add to timeline, split clip, add transition. Students observe.

Student Activity

Observe the demonstration. Identify timeline, preview, and tools. Note how transitions improve flow.

Explain (10-12 minutes)

Teacher Activity

Define video communication: sharing information through moving images and sound. File formats: MP4, AVI, MOV. Video creation process: idea → script → shoot → edit → export.

Student Activity

Take notes. Understand the video creation workflow. Compare different file formats.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Plan a 1-minute video about your school. Write the script, list shots needed, and plan transitions.' If no computer, create a storyboard on paper.

Student Activity

Work in groups. Write script, draw storyboard, plan transitions. Present plan to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What are the steps to create a video? 2. Name two video file formats. 3. Why are transitions important?'

Student Activity

Complete the exit ticket. Review the creation process.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Audio Editing with Audacity - network topology
Science: Scientific applications of Audio Editing with Audacity - data transmission
Language: Writing about Audio Editing with Audacity builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:
- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Audio Editing with Audacity concepts through practical exercises
Advanced: Apply Audio Editing with Audacity concepts to solve complex problems independently
SEND: Practice Audio Editing with Audacity skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Audio Editing with Audacity activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Audio Editing with Audacity into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.03 **Duration: 40 minutes**

Audio Mixing

Learning Objectives

- Import multiple audio tracks into Audacity
- Adjust volume levels and pan left/right channels
- Mix multiple tracks and export the final audio

Engage (5-7 minutes)

Teacher Activity

Show a short video clip 30 seconds. Ask: 'How do videos communicate messages better than text?' Discuss: visuals, audio, movement, emotion.

Student Activity

Watch the clip. Identify what makes it effective: music, transitions, text overlays. Share favorite video types.

Explore (10-12 minutes)

Teacher Activity

Show OpenShot interface or describe it. Demonstrate: import media, add to timeline, split clip, add transition. Students observe.

Student Activity

Observe the demonstration. Identify timeline, preview, and tools. Note how transitions improve flow.

Explain (10-12 minutes)

Teacher Activity

Define video communication: sharing information through moving images and sound. File formats: MP4, AVI, MOV. Video creation process: idea → script → shoot → edit → export.

Student Activity

Take notes. Understand the video creation workflow. Compare different file formats.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Plan a 1-minute video about your school. Write the script, list shots needed, and plan transitions.' If no computer, create a storyboard on paper.

Student Activity

Work in groups. Write script, draw storyboard, plan transitions. Present plan to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What are the steps to create a video? 2. Name two video file formats. 3. Why are transitions important?'

Student Activity

Complete the exit ticket. Review the creation process.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Audio Mixing - network topology
Science: Scientific applications of Audio Mixing - data transmission
Language: Writing about Audio Mixing builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Audio Mixing concepts through practical exercises
Advanced: Apply Audio Mixing concepts to solve complex problems independently
SEND: Practice Audio Mixing skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Audio Mixing activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Audio Mixing into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.04 **Duration: 40 minutes**

Video Concepts

Learning Objectives

- Define video and explain how frames create motion
- Identify video formats and their uses
- Describe the difference between analog and digital video

Engage (5-7 minutes)

Teacher Activity

Show a short video clip 30 seconds. Ask: 'How do videos communicate messages better than text?' Discuss: visuals, audio, movement, emotion.

Student Activity

Watch the clip. Identify what makes it effective: music, transitions, text overlays. Share favorite video types.

Explore (10-12 minutes)

Teacher Activity

Show OpenShot interface or describe it. Demonstrate: import media, add to timeline, split clip, add transition. Students observe.

Student Activity

Observe the demonstration. Identify timeline, preview, and tools. Note how transitions improve flow.

Explain (10-12 minutes)

Teacher Activity

Define video communication: sharing information through moving images and sound. File formats: MP4, AVI, MOV. Video creation process: idea → script → shoot → edit → export.

Student Activity

Take notes. Understand the video creation workflow. Compare different file formats.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Plan a 1-minute video about your school. Write the script, list shots needed, and plan transitions.' If no computer, create a storyboard on paper.

Student Activity

Work in groups. Write script, draw storyboard, plan transitions. Present plan to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What are the steps to create a video? 2. Name two video file formats. 3. Why are transitions important?'

Student Activity

Complete the exit ticket. Review the creation process.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Video Concepts - network topology
Science: Scientific applications of Video Concepts - data transmission
Language: Writing about Video Concepts builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:
- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Video Concepts concepts through practical exercises
Advanced: Apply Video Concepts concepts to solve complex problems independently
SEND: Practice Video Concepts skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Video Concepts activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Video Concepts into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.05 **Duration: 40 minutes**

Video Editing Basics

Learning Objectives

- Import video clips into a video editor
- Trim, split, and arrange clips on a timeline
- Add text overlays and export the edited video

Engage (5-7 minutes)

Teacher Activity

Show a short video clip 30 seconds. Ask: 'How do videos communicate messages better than text?' Discuss: visuals, audio, movement, emotion.

Student Activity

Watch the clip. Identify what makes it effective: music, transitions, text overlays. Share favorite video types.

Explore (10-12 minutes)

Teacher Activity

Show OpenShot interface or describe it. Demonstrate: import media, add to timeline, split clip, add transition. Students observe.

Student Activity

Observe the demonstration. Identify timeline, preview, and tools. Note how transitions improve flow.

Explain (10-12 minutes)

Teacher Activity

Define video communication: sharing information through moving images and sound. File formats: MP4, AVI, MOV. Video creation process: idea ⟶ script ⟶ shoot ⟶ edit ⟶ export.

Student Activity

Take notes. Understand the video creation workflow. Compare different file formats.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Plan a 1-minute video about your school. Write the script, list shots needed, and plan transitions.' If no computer, create a storyboard on paper.

Student Activity

Work in groups. Write script, draw storyboard, plan transitions. Present plan to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What are the steps to create a video? 2. Name two video file formats. 3. Why are transitions important?'

Student Activity

Complete the exit ticket. Review the creation process.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Video Editing Basics - network topology
Science: Scientific applications of Video Editing Basics - data transmission
Language: Writing about Video Editing Basics builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:
- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Video Editing Basics concepts through practical exercises
Advanced: Apply Video Editing Basics concepts to solve complex problems independently
SEND: Practice Video Editing Basics skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Video Editing Basics activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Video Editing Basics into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.06 **Duration: 40 minutes**

Presentation Software

Learning Objectives

- Identify features of presentation software like LibreOffice Impress
- Create a new presentation with multiple slides
- Add text, images, and shapes to slides

Engage (5-7 minutes)

Teacher Activity

Show a short video clip 30 seconds. Ask: ‘How do videos communicate messages better than text?’ Discuss: visuals, audio, movement, emotion.

Student Activity

Watch the clip. Identify what makes it effective: music, transitions, text overlays. Share favorite video types.

Explore (10-12 minutes)

Teacher Activity

Show OpenShot interface or describe it. Demonstrate: import media, add to timeline, split clip, add transition. Students observe.

Student Activity

Observe the demonstration. Identify timeline, preview, and tools. Note how transitions improve flow.

Explain (10-12 minutes)

Teacher Activity

Define video communication: sharing information through moving images and sound. File formats: MP4, AVI, MOV. Video creation process: idea ⟶ script ⟶ shoot ⟶ edit ⟶ export.

Student Activity

Take notes. Understand the video creation workflow. Compare different file formats.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Plan a 1-minute video about your school. Write the script, list shots needed, and plan transitions.’ If no computer, create a storyboard on paper.

Student Activity

Work in groups. Write script, draw storyboard, plan transitions. Present plan to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What are the steps to create a video? 2. Name two video file formats. 3. Why are transitions important?'

Student Activity

Complete the exit ticket. Review the creation process.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Presentation Software - network topology
Science: Scientific applications of Presentation Software - data transmission
Language: Writing about Presentation Software builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:
- Computer with presentation software
- design guide
- sample slides

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Presentation Software concepts through practical exercises
Advanced: Apply Presentation Software concepts to solve complex problems independently
SEND: Practice Presentation Software skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Presentation Software activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Presentation Software into manageable steps

Dormitory Task (Paper-and-Pencil)

Plan a 4-slide presentation about Presentation Software. Write the title and 3 bullet points for each slide.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.07 **Duration: 40 minutes**

Slide Design

Learning Objectives

- Apply consistent themes and backgrounds to slides
- Use fonts, colors, and alignment for readability
- Create visually appealing slide layouts

Engage (5-7 minutes)

Teacher Activity

Show a short video clip 30 seconds. Ask: 'How do videos communicate messages better than text?' Discuss: visuals, audio, movement, emotion.

Student Activity

Watch the clip. Identify what makes it effective: music, transitions, text overlays. Share favorite video types.

Explore (10-12 minutes)

Teacher Activity

Show OpenShot interface or describe it. Demonstrate: import media, add to timeline, split clip, add transition. Students observe.

Student Activity

Observe the demonstration. Identify timeline, preview, and tools. Note how transitions improve flow.

Explain (10-12 minutes)

Teacher Activity

Define video communication: sharing information through moving images and sound. File formats: MP4, AVI, MOV. Video creation process: idea ⟶ script ⟶ shoot ⟶ edit ⟶ export.

Student Activity

Take notes. Understand the video creation workflow. Compare different file formats.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Plan a 1-minute video about your school. Write the script, list shots needed, and plan transitions.' If no computer, create a storyboard on paper.

Student Activity

Work in groups. Write script, draw storyboard, plan transitions. Present plan to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What are the steps to create a video? 2. Name two video file formats. 3. Why are transitions important?'

Student Activity

Complete the exit ticket. Review the creation process.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Slide Design - network topology
Science: Scientific applications of Slide Design - data transmission
Language: Writing about Slide Design builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with presentation software
- design guide
- sample slides

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Slide Design concepts through practical exercises
Advanced: Apply Slide Design concepts to solve complex problems independently
SEND: Practice Slide Design skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Slide Design activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Slide Design into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Slide Design. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.08 **Duration: 40 minutes**

Animations

Learning Objectives

- Apply entrance, emphasis, and exit animations to objects
- Set animation order and timing
- Use animations effectively without overdoing them

Engage (5-7 minutes)

Teacher Activity

Show a short video clip 30 seconds. Ask: 'How do videos communicate messages better than text?' Discuss: visuals, audio, movement, emotion.

Student Activity

Watch the clip. Identify what makes it effective: music, transitions, text overlays. Share favorite video types.

Explore (10-12 minutes)

Teacher Activity

Show OpenShot interface or describe it. Demonstrate: import media, add to timeline, split clip, add transition. Students observe.

Student Activity

Observe the demonstration. Identify timeline, preview, and tools. Note how transitions improve flow.

Explain (10-12 minutes)

Teacher Activity

Define video communication: sharing information through moving images and sound. File formats: MP4, AVI, MOV. Video creation process: idea ⟶ script ⟶ shoot ⟶ edit ⟶ export.

Student Activity

Take notes. Understand the video creation workflow. Compare different file formats.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Plan a 1-minute video about your school. Write the script, list shots needed, and plan transitions.' If no computer, create a storyboard on paper.

Student Activity

Work in groups. Write script, draw storyboard, plan transitions. Present plan to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What are the steps to create a video? 2. Name two video file formats. 3. Why are transitions important?'

Student Activity

Complete the exit ticket. Review the creation process.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Animations - network topology
Science: Scientific applications of Animations - data transmission
Language: Writing about Animations builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:
- Networking reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Animations concepts through practical exercises
Advanced: Apply Animations concepts to solve complex problems independently
SEND: Practice Animations skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Animations activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Animations into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Animations. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.09 **Duration: 40 minutes**

Slide Transitions

Learning Objectives

- Apply different transition effects between slides
- Set transition duration and advance timing
- Choose appropriate transitions for different contexts

Engage (5-7 minutes)

Teacher Activity

Show a short video clip 30 seconds. Ask: 'How do videos communicate messages better than text?' Discuss: visuals, audio, movement, emotion.

Student Activity

Watch the clip. Identify what makes it effective: music, transitions, text overlays. Share favorite video types.

Explore (10-12 minutes)

Teacher Activity

Show OpenShot interface or describe it. Demonstrate: import media, add to timeline, split clip, add transition. Students observe.

Student Activity

Observe the demonstration. Identify timeline, preview, and tools. Note how transitions improve flow.

Explain (10-12 minutes)

Teacher Activity

Define video communication: sharing information through moving images and sound. File formats: MP4, AVI, MOV. Video creation process: idea ⟶ script ⟶ shoot ⟶ edit ⟶ export.

Student Activity

Take notes. Understand the video creation workflow. Compare different file formats.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Plan a 1-minute video about your school. Write the script, list shots needed, and plan transitions.' If no computer, create a storyboard on paper.

Student Activity

Work in groups. Write script, draw storyboard, plan transitions. Present plan to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What are the steps to create a video? 2. Name two video file formats. 3. Why are transitions important?'

Student Activity

Complete the exit ticket. Review the creation process.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Slide Transitions - network topology
Science: Scientific applications of Slide Transitions - data transmission
Language: Writing about Slide Transitions builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:
- Computer with presentation software
- design guide
- sample slides

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Slide Transitions concepts through practical exercises
Advanced: Apply Slide Transitions concepts to solve complex problems independently
SEND: Practice Slide Transitions skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Slide Transitions activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Slide Transitions into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Slide Transitions. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.10 **Duration: 40 minutes**

Multimedia Integration

Learning Objectives

- Embed audio and video clips into presentations
- Insert hyperlinks and interactive buttons
- Create a cohesive multimedia presentation

Engage (5-7 minutes)

Teacher Activity

Show a short video clip 30 seconds. Ask: 'How do videos communicate messages better than text?' Discuss: visuals, audio, movement, emotion.

Student Activity

Watch the clip. Identify what makes it effective: music, transitions, text overlays. Share favorite video types.

Explore (10-12 minutes)

Teacher Activity

Show OpenShot interface or describe it. Demonstrate: import media, add to timeline, split clip, add transition. Students observe.

Student Activity

Observe the demonstration. Identify timeline, preview, and tools. Note how transitions improve flow.

Explain (10-12 minutes)

Teacher Activity

Define video communication: sharing information through moving images and sound. File formats: MP4, AVI, MOV. Video creation process: idea ⟶ script ⟶ shoot ⟶ edit ⟶ export.

Student Activity

Take notes. Understand the video creation workflow. Compare different file formats.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Plan a 1-minute video about your school. Write the script, list shots needed, and plan transitions.' If no computer, create a storyboard on paper.

Student Activity

Work in groups. Write script, draw storyboard, plan transitions. Present plan to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What are the steps to create a video? 2. Name two video file formats. 3. Why are transitions important?'

Student Activity

Complete the exit ticket. Review the creation process.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Multimedia Integration - network topology
Science: Scientific applications of Multimedia Integration - data transmission
Language: Writing about Multimedia Integration builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Multimedia Integration concepts through practical exercises
Advanced: Apply Multimedia Integration concepts to solve complex problems independently
SEND: Practice Multimedia Integration skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Multimedia Integration activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Multimedia Integration into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Multimedia Integration. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.11 **Duration: 40 minutes**

Project - Presentation

Learning Objectives

- Plan and design a complete multimedia presentation
- Apply all learned skills: design, animation, transitions, and multimedia
- Present and evaluate peer presentations effectively

Engage (5-7 minutes)

Teacher Activity

Show a short video clip 30 seconds. Ask: ‘How do videos communicate messages better than text?’ Discuss: visuals, audio, movement, emotion.

Student Activity

Watch the clip. Identify what makes it effective: music, transitions, text overlays. Share favorite video types.

Explore (10-12 minutes)

Teacher Activity

Show OpenShot interface or describe it. Demonstrate: import media, add to timeline, split clip, add transition. Students observe.

Student Activity

Observe the demonstration. Identify timeline, preview, and tools. Note how transitions improve flow.

Explain (10-12 minutes)

Teacher Activity

Define video communication: sharing information through moving images and sound. File formats: MP4, AVI, MOV. Video creation process: idea ⟶ script ⟶ shoot ⟶ edit ⟶ export.

Student Activity

Take notes. Understand the video creation workflow. Compare different file formats.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Plan a 1-minute video about your school. Write the script, list shots needed, and plan transitions.’ If no computer, create a storyboard on paper.

Student Activity

Work in groups. Write script, draw storyboard, plan transitions. Present plan to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What are the steps to create a video? 2. Name two video file formats. 3. Why are transitions important?'

Student Activity

Complete the exit ticket. Review the creation process.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Presentation - network topology
Science: Scientific applications of Project - Presentation - data transmission
Language: Writing about Project - Presentation builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:
- Networking reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Project - Presentation concepts through practical exercises
Advanced: Apply Project - Presentation concepts to solve complex problems independently
SEND: Practice Project - Presentation skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment
CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Presentation project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Presentation project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.12 **Duration: 40 minutes**

Chapter 3 Review

Learning Objectives

- Recall all multimedia concepts from Chapter 3
- Demonstrate audio and video editing skills
- Apply presentation design and multimedia integration

Engage (5-7 minutes)

Teacher Activity

Show a short video clip 30 seconds. Ask: 'How do videos communicate messages better than text?' Discuss: visuals, audio, movement, emotion.

Student Activity

Watch the clip. Identify what makes it effective: music, transitions, text overlays. Share favorite video types.

Explore (10-12 minutes)

Teacher Activity

Show OpenShot interface or describe it. Demonstrate: import media, add to timeline, split clip, add transition. Students observe.

Student Activity

Observe the demonstration. Identify timeline, preview, and tools. Note how transitions improve flow.

Explain (10-12 minutes)

Teacher Activity

Define video communication: sharing information through moving images and sound. File formats: MP4, AVI, MOV. Video creation process: idea ⟶ script ⟶ shoot ⟶ edit ⟶ export.

Student Activity

Take notes. Understand the video creation workflow. Compare different file formats.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Plan a 1-minute video about your school. Write the script, list shots needed, and plan transitions.' If no computer, create a storyboard on paper.

Student Activity

Work in groups. Write script, draw storyboard, plan transitions. Present plan to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What are the steps to create a video? 2. Name two video file formats. 3. Why are transitions important?'

Student Activity

Complete the exit ticket. Review the creation process.

Cross-Curricular Connections

Mathematics: Review exercises consolidate database concepts
Science: Chapter summaries connect SQL topics
Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 3 Review skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 3 Review support
On Level: Demonstrate Chapter 3 Review through practical exercises
Advanced: Apply Chapter 3 Review to solve complex problems independently
SEND: Practice Chapter 3 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 3 Review activity
CT Skill Assessed: Decomposition - breaking down Chapter 3 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 3 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.01 **Duration: 40 minutes**

Introduction to Spreadsheets

Learning Objectives

- Define spreadsheet and its purpose in data management
- Identify rows, columns, cells, and cell references
- Open LibreOffice Calc and navigate the interface

Engage (5-7 minutes)

Teacher Activity

Show a poorly formatted document vs a professionally formatted one. Ask: 'Which would you submit as homework? Why?' Discuss formatting elements: alignment, fonts, spacing.

Student Activity

Compare both documents. Identify differences: alignment, headings, bullet points, spacing. Prefer the professional version.

Explore (10-12 minutes)

Teacher Activity

Open Writer. Demonstrate: alignment options, spell check, find/replace, bullets and numbering, inserting tables. Students follow along.

Student Activity

Follow along. Apply each formatting tool to a practice document. Experiment with different options.

Explain (10-12 minutes)

Teacher Activity

Explain document structuring: headings create hierarchy, bullets organize lists, tables present data. Show mail merge: 'How do you send the same letter to 40 students with different names?'

Student Activity

Take notes. Understand document structure. See how mail merge saves time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a class notice about a school event. Include: heading, date, details in bullet points, a table of schedules, and proper formatting.'

Student Activity

Work independently. Apply all formatting skills. Review against a checklist.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name three formatting tools. 2. What is mail merge used for? 3. Why is proper formatting important?'

Student Activity

Answer the questions. Check their document against the checklist.

Cross-Curricular Connections

Mathematics: Cells use row-column addresses like coordinates - math connection
Science: Spreadsheets organize data like scientific tables
Language: Writing clear data labels builds precise language

Digital Citizen Reminder: Do not enter personal information into shared spreadsheets.

Resources Needed:
- Computer with spreadsheet software
- sample data
- practice worksheet

Differentiation

Approaching: Enter text and numbers into cells
On Level: Create a simple data table with headers and data
Advanced: Design a complex spreadsheet with multiple data types and formatting
SEND: Enter data into cells with teacher guidance

Formative Assessment

Method: Data entry accuracy check
CT Skill Assessed: Decomposition - understanding rows, columns, and cells

Dormitory Task (Paper-and-Pencil)

Draw a grid with 5 columns and 6 rows on graph paper. Label the columns and fill in data about your class.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.02 **Duration: 40 minutes**

Basic Formulas

Learning Objectives

- Write formulas using addition, subtraction, multiplication, and division
- Use the equals sign to start formulas
- Understand the difference between formula and value

Engage (5-7 minutes)

Teacher Activity

Show a poorly formatted document vs a professionally formatted one. Ask: 'Which would you submit as homework? Why?' Discuss formatting elements: alignment, fonts, spacing.

Student Activity

Compare both documents. Identify differences: alignment, headings, bullet points, spacing. Prefer the professional version.

Explore (10-12 minutes)

Teacher Activity

Open Writer. Demonstrate: alignment options, spell check, find/replace, bullets and numbering, inserting tables. Students follow along.

Student Activity

Follow along. Apply each formatting tool to a practice document. Experiment with different options.

Explain (10-12 minutes)

Teacher Activity

Explain document structuring: headings create hierarchy, bullets organize lists, tables present data. Show mail merge: 'How do you send the same letter to 40 students with different names?'

Student Activity

Take notes. Understand document structure. See how mail merge saves time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a class notice about a school event. Include: heading, date, details in bullet points, a table of schedules, and proper formatting.'

Student Activity

Work independently. Apply all formatting skills. Review against a checklist.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name three formatting tools. 2. What is mail merge used for? 3. Why is proper formatting important?'

Student Activity

Answer the questions. Check their document against the checklist.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Basic Formulas - mathematical concepts
Science: Scientific applications of Basic Formulas - scientific applications
Language: Writing about Basic Formulas builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Computer with spreadsheet
- practice data set
- formula reference

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Basic Formulas concepts through practical exercises
Advanced: Apply Basic Formulas concepts to solve complex problems independently
SEND: Practice Basic Formulas skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Basic Formulas activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Basic Formulas into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Basic Formulas. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.03 **Duration: 40 minutes**

Cell References

Learning Objectives

- Distinguish between relative, absolute, and mixed references

Engage (5-7 minutes)

Teacher Activity

Show a poorly formatted document vs a professionally formatted one. Ask: 'Which would you submit as homework? Why?' Discuss formatting elements: alignment, fonts, spacing.

Student Activity

Compare both documents. Identify differences: alignment, headings, bullet points, spacing. Prefer the professional version.

Explore (10-12 minutes)

Teacher Activity

Open Writer. Demonstrate: alignment options, spell check, find/replace, bullets and numbering, inserting tables. Students follow along.

Student Activity

Follow along. Apply each formatting tool to a practice document. Experiment with different options.

Explain (10-12 minutes)

Teacher Activity

Explain document structuring: headings create hierarchy, bullets organize lists, tables present data. Show mail merge: 'How do you send the same letter to 40 students with different names?'

Student Activity

Take notes. Understand document structure. See how mail merge saves time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a class notice about a school event. Include: heading, date, details in bullet points, a table of schedules, and proper formatting.'

Student Activity

Work independently. Apply all formatting skills. Review against a checklist.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name three formatting tools. 2. What is mail merge used for? 3. Why is proper formatting important?'

Student Activity

Answer the questions. Check their document against the checklist.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Cell References - mathematical concepts
Science: Scientific applications of Cell References - scientific applications
Language: Writing about Cell References builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 4 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Cell References concepts through practical exercises
Advanced: Apply Cell References concepts to solve complex problems independently
SEND: Practice Cell References skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Cell References activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Cell References into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Cell References. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.04 **Duration: 40 minutes**

Functions

Learning Objectives

- Use SUM, AVERAGE, COUNT, MAX, and MIN functions
- Enter functions using the function wizard
- Apply functions to real-world data sets

Engage (5-7 minutes)

Teacher Activity

Show a poorly formatted document vs a professionally formatted one. Ask: 'Which would you submit as homework? Why?' Discuss formatting elements: alignment, fonts, spacing.

Student Activity

Compare both documents. Identify differences: alignment, headings, bullet points, spacing. Prefer the professional version.

Explore (10-12 minutes)

Teacher Activity

Open Writer. Demonstrate: alignment options, spell check, find/replace, bullets and numbering, inserting tables. Students follow along.

Student Activity

Follow along. Apply each formatting tool to a practice document. Experiment with different options.

Explain (10-12 minutes)

Teacher Activity

Explain document structuring: headings create hierarchy, bullets organize lists, tables present data. Show mail merge: 'How do you send the same letter to 40 students with different names?'

Student Activity

Take notes. Understand document structure. See how mail merge saves time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a class notice about a school event. Include: heading, date, details in bullet points, a table of schedules, and proper formatting.'

Student Activity

Work independently. Apply all formatting skills. Review against a checklist.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. Name three formatting tools. 2. What is mail merge used for? 3. Why is proper formatting important?’

Student Activity

Answer the questions. Check their document against the checklist.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Functions - mathematical concepts
Science: Scientific applications of Functions - scientific applications
Language: Writing about Functions builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Functions concepts through practical exercises
Advanced: Apply Functions concepts to solve complex problems independently
SEND: Practice Functions skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Functions activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Functions into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Functions. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.05 **Duration: 40 minutes**

Charts Creation

Learning Objectives

- Select data ranges for chart creation
- Create bar, line, and pie charts
- Choose appropriate chart types for different data

Engage (5-7 minutes)

Teacher Activity

Show a poorly formatted document vs a professionally formatted one. Ask: ‘Which would you submit as homework? Why?’ Discuss formatting elements: alignment, fonts, spacing.

Student Activity

Compare both documents. Identify differences: alignment, headings, bullet points, spacing. Prefer the professional version.

Explore (10-12 minutes)

Teacher Activity

Open Writer. Demonstrate: alignment options, spell check, find/replace, bullets and numbering, inserting tables. Students follow along.

Student Activity

Follow along. Apply each formatting tool to a practice document. Experiment with different options.

Explain (10-12 minutes)

Teacher Activity

Explain document structuring: headings create hierarchy, bullets organize lists, tables present data. Show mail merge: ‘How do you send the same letter to 40 students with different names?’

Student Activity

Take notes. Understand document structure. See how mail merge saves time.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Create a class notice about a school event. Include: heading, date, details in bullet points, a table of schedules, and proper formatting.’

Student Activity

Work independently. Apply all formatting skills. Review against a checklist.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name three formatting tools. 2. What is mail merge used for? 3. Why is proper formatting important?'

Student Activity

Answer the questions. Check their document against the checklist.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Charts Creation - mathematical concepts
Science: Scientific applications of Charts Creation - scientific applications
Language: Writing about Charts Creation builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Computer with spreadsheet
- practice data set
- formula reference

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Charts Creation concepts through practical exercises
Advanced: Apply Charts Creation concepts to solve complex problems independently
SEND: Practice Charts Creation skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Charts Creation activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Charts Creation into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Charts Creation. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.06 **Duration: 40 minutes**

Chart Formatting

Learning Objectives

- Modify chart titles, labels, and legends
- Change chart colors and styles
- Add trendlines and data labels to charts

Engage (5-7 minutes)

Teacher Activity

Show a poorly formatted document vs a professionally formatted one. Ask: 'Which would you submit as homework? Why?' Discuss formatting elements: alignment, fonts, spacing.

Student Activity

Compare both documents. Identify differences: alignment, headings, bullet points, spacing. Prefer the professional version.

Explore (10-12 minutes)

Teacher Activity

Open Writer. Demonstrate: alignment options, spell check, find/replace, bullets and numbering, inserting tables. Students follow along.

Student Activity

Follow along. Apply each formatting tool to a practice document. Experiment with different options.

Explain (10-12 minutes)

Teacher Activity

Explain document structuring: headings create hierarchy, bullets organize lists, tables present data. Show mail merge: 'How do you send the same letter to 40 students with different names?'

Student Activity

Take notes. Understand document structure. See how mail merge saves time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a class notice about a school event. Include: heading, date, details in bullet points, a table of schedules, and proper formatting.'

Student Activity

Work independently. Apply all formatting skills. Review against a checklist.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name three formatting tools. 2. What is mail merge used for? 3. Why is proper formatting important?'

Student Activity

Answer the questions. Check their document against the checklist.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Chart Formatting - mathematical concepts
Science: Scientific applications of Chart Formatting - scientific applications
Language: Writing about Chart Formatting builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Computer with spreadsheet
- practice data set
- formula reference

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Chart Formatting concepts through practical exercises
Advanced: Apply Chart Formatting concepts to solve complex problems independently
SEND: Practice Chart Formatting skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Chart Formatting activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Chart Formatting into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Chart Formatting. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.07 **Duration: 40 minutes**

Data Sorting

Learning Objectives

- Sort data in ascending and descending order
- Sort by single and multiple columns
- Use sort to organize records meaningfully

Engage (5-7 minutes)

Teacher Activity

Show a poorly formatted document vs a professionally formatted one. Ask: 'Which would you submit as homework? Why?' Discuss formatting elements: alignment, fonts, spacing.

Student Activity

Compare both documents. Identify differences: alignment, headings, bullet points, spacing. Prefer the professional version.

Explore (10-12 minutes)

Teacher Activity

Open Writer. Demonstrate: alignment options, spell check, find/replace, bullets and numbering, inserting tables. Students follow along.

Student Activity

Follow along. Apply each formatting tool to a practice document. Experiment with different options.

Explain (10-12 minutes)

Teacher Activity

Explain document structuring: headings create hierarchy, bullets organize lists, tables present data. Show mail merge: 'How do you send the same letter to 40 students with different names?'

Student Activity

Take notes. Understand document structure. See how mail merge saves time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a class notice about a school event. Include: heading, date, details in bullet points, a table of schedules, and proper formatting.'

Student Activity

Work independently. Apply all formatting skills. Review against a checklist.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name three formatting tools. 2. What is mail merge used for? 3. Why is proper formatting important?'

Student Activity

Answer the questions. Check their document against the checklist.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Data Sorting - mathematical concepts
Science: Scientific applications of Data Sorting - scientific applications
Language: Writing about Data Sorting builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 4 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Data Sorting concepts through practical exercises
Advanced: Apply Data Sorting concepts to solve complex problems independently
SEND: Practice Data Sorting skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Data Sorting activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Data Sorting into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Data Sorting. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.08 **Duration: 40 minutes**

Data Filtering

Learning Objectives

- Apply auto-filter to display specific records
- Use filter criteria to show data meeting conditions
- Clear filters and understand filtered vs hidden data

Engage (5-7 minutes)

Teacher Activity

Show a poorly formatted document vs a professionally formatted one. Ask: 'Which would you submit as homework? Why?' Discuss formatting elements: alignment, fonts, spacing.

Student Activity

Compare both documents. Identify differences: alignment, headings, bullet points, spacing. Prefer the professional version.

Explore (10-12 minutes)

Teacher Activity

Open Writer. Demonstrate: alignment options, spell check, find/replace, bullets and numbering, inserting tables. Students follow along.

Student Activity

Follow along. Apply each formatting tool to a practice document. Experiment with different options.

Explain (10-12 minutes)

Teacher Activity

Explain document structuring: headings create hierarchy, bullets organize lists, tables present data. Show mail merge: 'How do you send the same letter to 40 students with different names?'

Student Activity

Take notes. Understand document structure. See how mail merge saves time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a class notice about a school event. Include: heading, date, details in bullet points, a table of schedules, and proper formatting.'

Student Activity

Work independently. Apply all formatting skills. Review against a checklist.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name three formatting tools. 2. What is mail merge used for? 3. Why is proper formatting important?'

Student Activity

Answer the questions. Check their document against the checklist.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Data Filtering - mathematical concepts
Science: Scientific applications of Data Filtering - scientific applications
Language: Writing about Data Filtering builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Data Filtering concepts through practical exercises
Advanced: Apply Data Filtering concepts to solve complex problems independently
SEND: Practice Data Filtering skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Data Filtering activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Data Filtering into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Data Filtering. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.09 **Duration: 40 minutes**

Data Validation

Learning Objectives

- Set validation rules for data entry in cells
- Create dropdown lists for restricted choices
- Display error messages for invalid entries

Engage (5-7 minutes)

Teacher Activity

Show a poorly formatted document vs a professionally formatted one. Ask: 'Which would you submit as homework? Why?' Discuss formatting elements: alignment, fonts, spacing.

Student Activity

Compare both documents. Identify differences: alignment, headings, bullet points, spacing. Prefer the professional version.

Explore (10-12 minutes)

Teacher Activity

Open Writer. Demonstrate: alignment options, spell check, find/replace, bullets and numbering, inserting tables. Students follow along.

Student Activity

Follow along. Apply each formatting tool to a practice document. Experiment with different options.

Explain (10-12 minutes)

Teacher Activity

Explain document structuring: headings create hierarchy, bullets organize lists, tables present data. Show mail merge: 'How do you send the same letter to 40 students with different names?'

Student Activity

Take notes. Understand document structure. See how mail merge saves time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a class notice about a school event. Include: heading, date, details in bullet points, a table of schedules, and proper formatting.'

Student Activity

Work independently. Apply all formatting skills. Review against a checklist.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name three formatting tools. 2. What is mail merge used for? 3. Why is proper formatting important?'

Student Activity

Answer the questions. Check their document against the checklist.

Cross-Curricular Connections

Mathematics: Data validation uses rules and constraints - logical conditions
Science: Error prevention through validation - quality control
Language: Writing validation rules builds precise specification skills

Digital Citizen Reminder: Data validation prevents errors - always set up rules for data entry.

Resources Needed:
- Spreadsheet
- validation examples
- practice exercises

Differentiation

Approaching: Add a dropdown list to a cell
On Level: Create data validation with custom error messages
Advanced: Design comprehensive validation rules for a complete data entry system
SEND: Add a dropdown list with teacher guidance

Formative Assessment

Method: Data validation setup task
CT Skill Assessed: Decomposition - breaking validation rules into specific conditions

Dormitory Task (Paper-and-Pencil)

Write three data validation rules for a student marks spreadsheet. Example: Marks must be between 0 and 100.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.10 **Duration: 40 minutes**

Conditional Formatting

Learning Objectives

- Apply color scales to visualize data ranges
- Use data bars and icon sets for quick analysis
- Create custom rules for highlighting specific cells

Engage (5-7 minutes)

Teacher Activity

Show a poorly formatted document vs a professionally formatted one. Ask: 'Which would you submit as homework? Why?' Discuss formatting elements: alignment, fonts, spacing.

Student Activity

Compare both documents. Identify differences: alignment, headings, bullet points, spacing. Prefer the professional version.

Explore (10-12 minutes)

Teacher Activity

Open Writer. Demonstrate: alignment options, spell check, find/replace, bullets and numbering, inserting tables. Students follow along.

Student Activity

Follow along. Apply each formatting tool to a practice document. Experiment with different options.

Explain (10-12 minutes)

Teacher Activity

Explain document structuring: headings create hierarchy, bullets organize lists, tables present data. Show mail merge: 'How do you send the same letter to 40 students with different names?'

Student Activity

Take notes. Understand document structure. See how mail merge saves time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a class notice about a school event. Include: heading, date, details in bullet points, a table of schedules, and proper formatting.'

Student Activity

Work independently. Apply all formatting skills. Review against a checklist.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name three formatting tools. 2. What is mail merge used for? 3. Why is proper formatting important?'

Student Activity

Answer the questions. Check their document against the checklist.

Cross-Curricular Connections

Mathematics: IF conditions use logical operators (>, <, =) - Boolean math
Science: Conditional formatting highlights patterns in data
Language: Writing IF conditions builds logical thinking skills

Digital Citizen Reminder: Use conditional formatting to highlight important data, not to hide it.

Resources Needed:

- Spreadsheet with data
- condition examples
- practice exercises

Differentiation

Approaching: Apply color to cells that meet one condition
On Level: Use IF function to display different text based on values
Advanced: Apply multiple conditional formatting rules with complex conditions
SEND: Apply a simple condition with teacher guidance

Formative Assessment

Method: Conditional formatting task
CT Skill Assessed: Pattern Recognition - identifying data patterns through visual highlighting

Dormitory Task (Paper-and-Pencil)

Write 5 conditions and their outcomes. Example: If score > 90, display Excellent.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.11 **Duration: 40 minutes**

Project - Grade Sheet

Learning Objectives

- Create a complete grade sheet using all spreadsheet skills
- Apply formulas, functions, charts, and formatting
- Analyze and present data findings effectively

Engage (5-7 minutes)

Teacher Activity

Show a poorly formatted document vs a professionally formatted one. Ask: 'Which would you submit as homework? Why?' Discuss formatting elements: alignment, fonts, spacing.

Student Activity

Compare both documents. Identify differences: alignment, headings, bullet points, spacing. Prefer the professional version.

Explore (10-12 minutes)

Teacher Activity

Open Writer. Demonstrate: alignment options, spell check, find/replace, bullets and numbering, inserting tables. Students follow along.

Student Activity

Follow along. Apply each formatting tool to a practice document. Experiment with different options.

Explain (10-12 minutes)

Teacher Activity

Explain document structuring: headings create hierarchy, bullets organize lists, tables present data. Show mail merge: 'How do you send the same letter to 40 students with different names?'

Student Activity

Take notes. Understand document structure. See how mail merge saves time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a class notice about a school event. Include: heading, date, details in bullet points, a table of schedules, and proper formatting.'

Student Activity

Work independently. Apply all formatting skills. Review against a checklist.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name three formatting tools. 2. What is mail merge used for? 3. Why is proper formatting important?'

Student Activity

Answer the questions. Check their document against the checklist.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Grade Sheet - mathematical concepts
Science: Scientific applications of Project - Grade Sheet - scientific applications
Language: Writing about Project - Grade Sheet builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 4 reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Project - Grade Sheet concepts through practical exercises
Advanced: Apply Project - Grade Sheet concepts to solve complex problems independently
SEND: Practice Project - Grade Sheet skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment
CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Grade Sheet project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Grade Sheet project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.12 **Duration: 40 minutes**

Chapter 4 Review

Learning Objectives

- Recall all data handling concepts from Chapter 4
- Demonstrate proficiency in formulas, functions, and charts
- Apply data analysis skills to solve real-world problems

Engage (5-7 minutes)

Teacher Activity

Show a poorly formatted document vs a professionally formatted one. Ask: 'Which would you submit as homework? Why?' Discuss formatting elements: alignment, fonts, spacing.

Student Activity

Compare both documents. Identify differences: alignment, headings, bullet points, spacing. Prefer the professional version.

Explore (10-12 minutes)

Teacher Activity

Open Writer. Demonstrate: alignment options, spell check, find/replace, bullets and numbering, inserting tables. Students follow along.

Student Activity

Follow along. Apply each formatting tool to a practice document. Experiment with different options.

Explain (10-12 minutes)

Teacher Activity

Explain document structuring: headings create hierarchy, bullets organize lists, tables present data. Show mail merge: 'How do you send the same letter to 40 students with different names?'

Student Activity

Take notes. Understand document structure. See how mail merge saves time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a class notice about a school event. Include: heading, date, details in bullet points, a table of schedules, and proper formatting.'

Student Activity

Work independently. Apply all formatting skills. Review against a checklist.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name three formatting tools. 2. What is mail merge used for? 3. Why is proper formatting important?'

Student Activity

Answer the questions. Check their document against the checklist.

Cross-Curricular Connections

Mathematics: Review exercises consolidate web development concepts
Science: Chapter summaries connect HTML and CSS topics
Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 4 Review skills responsibly and ethically.

Resources Needed:
- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 4 Review support
On Level: Demonstrate Chapter 4 Review through practical exercises
Advanced: Apply Chapter 4 Review to solve complex problems independently
SEND: Practice Chapter 4 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 4 Review activity
CT Skill Assessed: Decomposition - breaking down Chapter 4 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 4 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.01 **Duration: 40 minutes**

Scratch Review

Learning Objectives

- Recall Scratch interface components and their functions
- Identify sprites, costumes, and backdrops in projects
- Navigate the Scratch programming environment confidently

Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow keys? How does the score increase?' Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

Explain (10-12 minutes)

Teacher Activity

Review Scratch concepts: events when key pressed, loops forever, repeat, conditionals if touching, variables score. Introduce My Blocks: creating custom functions.

Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.' Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Scratch Review - mathematical concepts
Science: Scientific applications of Scratch Review - scientific applications
Language: Writing about Scratch Review builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Chapter 5 reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Scratch Review concepts through practical exercises
Advanced: Apply Scratch Review concepts to solve complex problems independently
SEND: Practice Scratch Review skills with step-by-step teacher guidance

Formative Assessment

Method: Chapter review quiz and self-assessment
CT Skill Assessed: Evaluation - self-assessment of Chapter 5 learning

Dormitory Task (Paper-and-Pencil)

Write a summary of everything you learned about Chapter 5. Include at least 8 key points.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.02 **Duration: 40 minutes**

Variables Advanced

Learning Objectives

- Create and use variables in Scratch projects
- Understand the difference between local and global variables
- Implement variables for score tracking and game mechanics

Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: ‘What happens when you press arrow keys? How does the score increase?’ Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

Explain (10-12 minutes)

Teacher Activity

Review Scratch concepts: events when key pressed, loops forever, repeat, conditionals if touching, variables score. Introduce My Blocks: creating custom functions.

Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.’ Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Variables Advanced - mathematical concepts
Science: Scientific applications of Variables Advanced - scientific applications
Language: Writing about Variables Advanced builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Variables Advanced concepts through practical exercises
Advanced: Apply Variables Advanced concepts to solve complex problems independently
SEND: Practice Variables Advanced skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Variables Advanced activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Variables Advanced into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Variables Advanced. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.03 **Duration: 40 minutes**

Conditional Logic

Learning Objectives

- Understand if-then and if-then-else blocks in Scratch
- Create programs that make decisions based on conditions
- Apply conditional logic to real-world problem scenarios

Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow keys? How does the score increase?' Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

Explain (10-12 minutes)

Teacher Activity

Review Scratch concepts: events when key pressed, loops forever, repeat, conditionals if touching, variables score. Introduce My Blocks: creating custom functions.

Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.' Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Conditional Logic - mathematical concepts
Science: Scientific applications of Conditional Logic - scientific applications
Language: Writing about Conditional Logic builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Conditional Logic concepts through practical exercises
Advanced: Apply Conditional Logic concepts to solve complex problems independently
SEND: Practice Conditional Logic skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Conditional Logic activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Conditional Logic into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Conditional Logic. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.04 **Duration: 40 minutes**

Loops Advanced

Learning Objectives

- Use repeat, forever, and repeat-until loops in Scratch
- Understand when to use different loop types
- Create animated sequences using loops effectively

Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow keys? How does the score increase?' Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

Explain (10-12 minutes)

Teacher Activity

Review Scratch concepts: events when key pressed, loops forever, repeat, conditionals if touching, variables score. Introduce My Blocks: creating custom functions.

Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.' Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Loops Advanced - mathematical concepts
Science: Scientific applications of Loops Advanced - scientific applications
Language: Writing about Loops Advanced builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Loops Advanced concepts through practical exercises
Advanced: Apply Loops Advanced concepts to solve complex problems independently
SEND: Practice Loops Advanced skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Loops Advanced activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Loops Advanced into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Loops Advanced. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.05 **Duration: 40 minutes**

Sensing Blocks

Learning Objectives

- Explore Scratch sensing blocks for input detection
- Use distance, color, and key-press sensors in programs
- Create interactive projects that respond to user actions

Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow keys? How does the score increase?' Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

Explain (10-12 minutes)

Teacher Activity

Review Scratch concepts: events when key pressed, loops forever, repeat, conditionals if touching, variables score. Introduce My Blocks: creating custom functions.

Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.' Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Sensing Blocks - mathematical concepts
Science: Scientific applications of Sensing Blocks - scientific applications
Language: Writing about Sensing Blocks builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Computer with Scratch
- block reference card
- project worksheet

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Sensing Blocks concepts through practical exercises
Advanced: Apply Sensing Blocks concepts to solve complex problems independently
SEND: Practice Sensing Blocks skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Sensing Blocks activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Sensing Blocks into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Sensing Blocks. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.06 **Duration: 40 minutes**

Sprite Interactions

Learning Objectives

- Create multiple sprites that interact with each other
- Use broadcasts to coordinate sprite communication
- Design scenarios where sprites work together to complete tasks

Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow keys? How does the score increase?' Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

Explain (10-12 minutes)

Teacher Activity

Review Scratch concepts: events when key pressed, loops forever, repeat, conditionals if touching, variables score. Introduce My Blocks: creating custom functions.

Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.' Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Sprite Interactions - mathematical concepts
Science: Scientific applications of Sprite Interactions - scientific applications
Language: Writing about Sprite Interactions builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Sprite Interactions concepts through practical exercises
Advanced: Apply Sprite Interactions concepts to solve complex problems independently
SEND: Practice Sprite Interactions skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Sprite Interactions activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Sprite Interactions into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Sprite Interactions. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.07 **Duration: 40 minutes**

Clones

Learning Objectives

- Understand what clones are and how they work in Scratch
- Create and delete clones dynamically during program execution
- Use clones to build multi-character game scenarios

Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: ‘What happens when you press arrow keys? How does the score increase?’ Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

Explain (10-12 minutes)

Teacher Activity

Review Scratch concepts: events when key pressed, loops forever, repeat, conditionals if touching, variables score. Introduce My Blocks: creating custom functions.

Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.’ Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Clones - mathematical concepts
Science: Scientific applications of Clones - scientific applications
Language: Writing about Clones builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Clones concepts through practical exercises
Advanced: Apply Clones concepts to solve complex problems independently
SEND: Practice Clones skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Clones activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Clones into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Clones. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.08 | **Duration: 40 minutes**

Game Design

Learning Objectives

- Understand core game mechanics: objectives, rules, and feedback
- Apply game design principles to create engaging gameplay
- Balance difficulty and reward in game elements

Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow keys? How does the score increase?' Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

Explain (10-12 minutes)

Teacher Activity

Review Scratch concepts: events when key pressed, loops forever, repeat, conditionals if touching, variables score. Introduce My Blocks: creating custom functions.

Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.' Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Game Design - mathematical concepts
Science: Scientific applications of Game Design - scientific applications
Language: Writing about Game Design builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Game Design concepts through practical exercises
Advanced: Apply Game Design concepts to solve complex problems independently
SEND: Practice Game Design skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Game Design activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Game Design into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Game Design. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.09 | **Duration: 40 minutes**

Keyboard Controls

Learning Objectives

- Implement keyboard input controls for sprite movement
- Create responsive controls using arrow keys and other inputs
- Design smooth movement systems for game characters

Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow keys? How does the score increase?' Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

Explain (10-12 minutes)

Teacher Activity

Review Scratch concepts: events when key pressed, loops forever, repeat, conditionals if touching, variables score. Introduce My Blocks: creating custom functions.

Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.' Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Keyboard Controls - mathematical concepts
Science: Scientific applications of Keyboard Controls - scientific applications
Language: Writing about Keyboard Controls builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Keyboard Controls concepts through practical exercises
Advanced: Apply Keyboard Controls concepts to solve complex problems independently
SEND: Practice Keyboard Controls skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Keyboard Controls activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Keyboard Controls into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Keyboard Controls. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.10 **Duration: 40 minutes**

Random Events

Learning Objectives

- Use random number generation in Scratch programs
- Create unpredictable game elements using randomness
- Understand how randomness adds variety and replayability

Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow keys? How does the score increase?' Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

Explain (10-12 minutes)

Teacher Activity

Review Scratch concepts: events when key pressed, loops forever, repeat, conditionals if touching, variables score. Introduce My Blocks: creating custom functions.

Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.' Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Random Events - mathematical concepts
Science: Scientific applications of Random Events - scientific applications
Language: Writing about Random Events builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Random Events concepts through practical exercises
Advanced: Apply Random Events concepts to solve complex problems independently
SEND: Practice Random Events skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Random Events activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Random Events into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Random Events. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.11 **Duration: 40 minutes**

Project - Game

Learning Objectives

- Apply all learned Scratch concepts to create a complete game
- Integrate variables, loops, conditionals, and sprite interactions
- Document and present a finished game project to peers

Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: 'What happens when you press arrow keys? How does the score increase?' Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

Explain (10-12 minutes)

Teacher Activity

Review Scratch concepts: events when key pressed, loops forever, repeat, conditionals if touching, variables score. Introduce My Blocks: creating custom functions.

Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.' Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Game - mathematical concepts
Science: Scientific applications of Project - Game - scientific applications
Language: Writing about Project - Game builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Chapter 5 reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Project - Game concepts through practical exercises
Advanced: Apply Project - Game concepts to solve complex problems independently
SEND: Practice Project - Game skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment
CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Game project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Game project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.12 **Duration: 40 minutes**

Chapter 5 Review

Learning Objectives

- Review and consolidate all Scratch concepts from Chapter 5
- Demonstrate mastery through a comprehensive review activity
- Identify areas for further practice and improvement

Engage (5-7 minutes)

Teacher Activity

Show a Scratch game Catch the Apples. Ask: ‘What happens when you press arrow keys? How does the score increase?’ Discuss the logic behind the game.

Student Activity

Play the game briefly. Observe: sprite moves, apples fall, score increases. Guess how each part works.

Explore (10-12 minutes)

Teacher Activity

Open the game project in Scratch. Show the scripts for each sprite. Let students experiment by changing values speed, score.

Student Activity

Explore the scripts. Change the move speed from 10 to 20. Observe the difference. Try adding a new block.

Explain (10-12 minutes)

Teacher Activity

Review Scratch concepts: events when key pressed, loops forever, repeat, conditionals if touching, variables score. Introduce My Blocks: creating custom functions.

Student Activity

Take notes. Understand how events, loops, and variables work together. Practice creating a My Block.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Add a new feature to the game — a timer that counts down from 60 seconds. When it reaches zero, the game ends.’ Guide through the logic.

Student Activity

Work in pairs. Create a timer variable. Add countdown logic. Test and debug.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What event starts the game? 2. How does the score variable increase? 3. What is a My Block?'

Student Activity

Answer the questions. Show their timer feature to a partner.

Cross-Curricular Connections

Mathematics: Review exercises consolidate algorithm concepts
Science: Chapter summaries connect all algorithm topics
Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 5 Review skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 5 Review support
On Level: Demonstrate Chapter 5 Review through practical exercises
Advanced: Apply Chapter 5 Review to solve complex problems independently
SEND: Practice Chapter 5 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 5 Review activity
CT Skill Assessed: Decomposition - breaking down Chapter 5 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 5 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.01 **Duration: 40 minutes**

What is Digital Citizenship?

Learning Objectives

- Define digital citizenship and its importance in modern life
- Identify the core principles of being a responsible digital citizen
- Connect digital citizenship concepts to real-life village community values

Engage (5-7 minutes)

Teacher Activity

Ask: 'What software do you use every day? What would happen if all software disappeared?' Discuss how software makes computers useful.

Student Activity

List software: browser, WhatsApp, games, calculator. Discuss dependency on software for daily tasks.

Explore (10-12 minutes)

Teacher Activity

Show categories of software: system OS, drivers, application browser, editor, utility antivirus, backup. Students categorize given examples.

Student Activity

Work in pairs. Sort 15 software names into correct categories. Discuss any disagreements.

Explain (10-12 minutes)

Teacher Activity

Define each category with examples: OS manages hardware, applications do specific tasks, utilities maintain the system. Show Linux as a free OS alternative.

Student Activity

Take notes. Understand the hierarchy: hardware ⟶ OS ⟶ applications. See Linux as an option.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a software guide for a new student. List 5 essential software for school work, explain what each does, and where to get it.'

Student Activity

Research and write the guide. Include free alternatives where possible. Format as a handout.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name one system software and one application software. 2. What is the difference between free and open-source software? 3. Why is antivirus important?'

Student Activity

Complete the exit ticket. Review software categories.

Cross-Curricular Connections

Mathematics: Mathematical concepts in What is Digital Citizenship? - mathematical concepts
Science: Scientific applications of What is Digital Citizenship? - scientific applications
Language: Writing about What is Digital Citizenship? builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Scenario cards
- discussion guide
- pledge template

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of What is Digital Citizenship? concepts through practical exercises
Advanced: Apply What is Digital Citizenship? concepts to solve complex problems independently
SEND: Practice What is Digital Citizenship? skills with step-by-step teacher guidance

Formative Assessment

Method: Practical What is Digital Citizenship? activity and oral questioning
CT Skill Assessed: Decomposition - breaking down What is Digital Citizenship? into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 rules for being a responsible digital citizen. Explain why each rule is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.02 **Duration: 40 minutes**

Online Safety Rules

Learning Objectives

- Learn essential online safety rules for internet use
- Identify common online dangers and how to avoid them
- Create a personal online safety checklist

Engage (5-7 minutes)

Teacher Activity

Ask: 'What software do you use every day? What would happen if all software disappeared?' Discuss how software makes computers useful.

Student Activity

List software: browser, WhatsApp, games, calculator. Discuss dependency on software for daily tasks.

Explore (10-12 minutes)

Teacher Activity

Show categories of software: system OS, drivers, application browser, editor, utility antivirus, backup. Students categorize given examples.

Student Activity

Work in pairs. Sort 15 software names into correct categories. Discuss any disagreements.

Explain (10-12 minutes)

Teacher Activity

Define each category with examples: OS manages hardware, applications do specific tasks, utilities maintain the system. Show Linux as a free OS alternative.

Student Activity

Take notes. Understand the hierarchy: hardware → OS → applications. See Linux as an option.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a software guide for a new student. List 5 essential software for school work, explain what each does, and where to get it.'

Student Activity

Research and write the guide. Include free alternatives where possible. Format as a handout.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name one system software and one application software. 2. What is the difference between free and open-source software? 3. Why is antivirus important?'

Student Activity

Complete the exit ticket. Review software categories.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Online Safety Rules - mathematical concepts
Science: Scientific applications of Online Safety Rules - scientific applications
Language: Writing about Online Safety Rules builds technical vocabulary and communication skills

Digital Citizen Reminder: Follow all online safety rules to protect yourself and your information.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Online Safety Rules concepts through practical exercises
Advanced: Apply Online Safety Rules concepts to solve complex problems independently
SEND: Practice Online Safety Rules skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Online Safety Rules activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Online Safety Rules into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Online Safety Rules. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.03 **Duration: 40 minutes**

Password Security

Learning Objectives

- Understand why passwords are important for online security
- Learn to create strong, memorable passwords
- Practice good password management habits

Engage (5-7 minutes)

Teacher Activity

Ask: 'What software do you use every day? What would happen if all software disappeared?' Discuss how software makes computers useful.

Student Activity

List software: browser, WhatsApp, games, calculator. Discuss dependency on software for daily tasks.

Explore (10-12 minutes)

Teacher Activity

Show categories of software: system OS, drivers, application browser, editor, utility antivirus, backup. Students categorize given examples.

Student Activity

Work in pairs. Sort 15 software names into correct categories. Discuss any disagreements.

Explain (10-12 minutes)

Teacher Activity

Define each category with examples: OS manages hardware, applications do specific tasks, utilities maintain the system. Show Linux as a free OS alternative.

Student Activity

Take notes. Understand the hierarchy: hardware → OS → applications. See Linux as an option.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a software guide for a new student. List 5 essential software for school work, explain what each does, and where to get it.'

Student Activity

Research and write the guide. Include free alternatives where possible. Format as a handout.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name one system software and one application software. 2. What is the difference between free and open-source software? 3. Why is antivirus important?'

Student Activity

Complete the exit ticket. Review software categories.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Password Security - mathematical concepts
Science: Scientific applications of Password Security - scientific applications
Language: Writing about Password Security builds technical vocabulary and communication skills

Digital Citizen Reminder: Use strong passwords with at least 8 characters including letters, numbers, and symbols.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Password Security concepts through practical exercises
Advanced: Apply Password Security concepts to solve complex problems independently
SEND: Practice Password Security skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Password Security activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Password Security into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Password Security. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.04 **Duration: 40 minutes**

Cyberbullying

Learning Objectives

- Define cyberbullying and recognize its various forms
- Understand the emotional and social impact of cyberbullying
- Learn strategies to prevent and respond to cyberbullying

Engage (5-7 minutes)

Teacher Activity

Ask: 'What software do you use every day? What would happen if all software disappeared?' Discuss how software makes computers useful.

Student Activity

List software: browser, WhatsApp, games, calculator. Discuss dependency on software for daily tasks.

Explore (10-12 minutes)

Teacher Activity

Show categories of software: system OS, drivers, application browser, editor, utility antivirus, backup. Students categorize given examples.

Student Activity

Work in pairs. Sort 15 software names into correct categories. Discuss any disagreements.

Explain (10-12 minutes)

Teacher Activity

Define each category with examples: OS manages hardware, applications do specific tasks, utilities maintain the system. Show Linux as a free OS alternative.

Student Activity

Take notes. Understand the hierarchy: hardware → OS → applications. See Linux as an option.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a software guide for a new student. List 5 essential software for school work, explain what each does, and where to get it.'

Student Activity

Research and write the guide. Include free alternatives where possible. Format as a handout.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name one system software and one application software. 2. What is the difference between free and open-source software? 3. Why is antivirus important?'

Student Activity

Complete the exit ticket. Review software categories.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Cyberbullying - mathematical concepts
Science: Scientific applications of Cyberbullying - scientific applications
Language: Writing about Cyberbullying builds technical vocabulary and communication skills

Digital Citizen Reminder: Never cyberbully others - treat everyone with respect online.

Resources Needed:
- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Cyberbullying concepts through practical exercises
Advanced: Apply Cyberbullying concepts to solve complex problems independently
SEND: Practice Cyberbullying skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Cyberbullying activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Cyberbullying into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Cyberbullying. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.05 **Duration: 40 minutes**

Digital Footprint

Learning Objectives

- Understand what a digital footprint is and how it's created
- Recognize the permanence of online actions and their consequences
- Learn to manage and protect their digital footprint

Engage (5-7 minutes)

Teacher Activity

Ask: 'What software do you use every day? What would happen if all software disappeared?' Discuss how software makes computers useful.

Student Activity

List software: browser, WhatsApp, games, calculator. Discuss dependency on software for daily tasks.

Explore (10-12 minutes)

Teacher Activity

Show categories of software: system OS, drivers, application browser, editor, utility antivirus, backup. Students categorize given examples.

Student Activity

Work in pairs. Sort 15 software names into correct categories. Discuss any disagreements.

Explain (10-12 minutes)

Teacher Activity

Define each category with examples: OS manages hardware, applications do specific tasks, utilities maintain the system. Show Linux as a free OS alternative.

Student Activity

Take notes. Understand the hierarchy: hardware ⟶ OS ⟶ applications. See Linux as an option.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a software guide for a new student. List 5 essential software for school work, explain what each does, and where to get it.'

Student Activity

Research and write the guide. Include free alternatives where possible. Format as a handout.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name one system software and one application software. 2. What is the difference between free and open-source software? 3. Why is antivirus important?'

Student Activity

Complete the exit ticket. Review software categories.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Digital Footprint - mathematical concepts
Science: Scientific applications of Digital Footprint - scientific applications
Language: Writing about Digital Footprint builds technical vocabulary and communication skills

Digital Citizen Reminder: Your digital footprint lasts forever - think before you post.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Digital Footprint concepts through practical exercises
Advanced: Apply Digital Footprint concepts to solve complex problems independently
SEND: Practice Digital Footprint skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Digital Footprint activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Digital Footprint into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Digital Footprint. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.06 **Duration: 40 minutes**

Online Predators

Learning Objectives

- Understand who online predators are and how they target children
- Recognize warning signs of predatory behavior online
- Learn protective strategies to stay safe from online predators

Engage (5-7 minutes)

Teacher Activity

Ask: 'What software do you use every day? What would happen if all software disappeared?' Discuss how software makes computers useful.

Student Activity

List software: browser, WhatsApp, games, calculator. Discuss dependency on software for daily tasks.

Explore (10-12 minutes)

Teacher Activity

Show categories of software: system OS, drivers, application browser, editor, utility antivirus, backup. Students categorize given examples.

Student Activity

Work in pairs. Sort 15 software names into correct categories. Discuss any disagreements.

Explain (10-12 minutes)

Teacher Activity

Define each category with examples: OS manages hardware, applications do specific tasks, utilities maintain the system. Show Linux as a free OS alternative.

Student Activity

Take notes. Understand the hierarchy: hardware → OS → applications. See Linux as an option.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a software guide for a new student. List 5 essential software for school work, explain what each does, and where to get it.'

Student Activity

Research and write the guide. Include free alternatives where possible. Format as a handout.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name one system software and one application software. 2. What is the difference between free and open-source software? 3. Why is antivirus important?'

Student Activity

Complete the exit ticket. Review software categories.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Online Predators - mathematical concepts
Science: Scientific applications of Online Predators - scientific applications
Language: Writing about Online Predators builds technical vocabulary and communication skills

Digital Citizen Reminder: Never meet online strangers in person - always tell a parent or teacher.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Online Predators concepts through practical exercises
Advanced: Apply Online Predators concepts to solve complex problems independently
SEND: Practice Online Predators skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Online Predators activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Online Predators into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Online Predators. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.07 **Duration: 40 minutes**

Copyright Basics

Learning Objectives

- Understand what copyright is and why it exists
- Recognize copyright symbols and their meaning
- Learn how to use copyrighted material legally

Engage (5-7 minutes)

Teacher Activity

Ask: 'What software do you use every day? What would happen if all software disappeared?' Discuss how software makes computers useful.

Student Activity

List software: browser, WhatsApp, games, calculator. Discuss dependency on software for daily tasks.

Explore (10-12 minutes)

Teacher Activity

Show categories of software: system OS, drivers, application browser, editor, utility antivirus, backup. Students categorize given examples.

Student Activity

Work in pairs. Sort 15 software names into correct categories. Discuss any disagreements.

Explain (10-12 minutes)

Teacher Activity

Define each category with examples: OS manages hardware, applications do specific tasks, utilities maintain the system. Show Linux as a free OS alternative.

Student Activity

Take notes. Understand the hierarchy: hardware ⟶ OS ⟶ applications. See Linux as an option.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a software guide for a new student. List 5 essential software for school work, explain what each does, and where to get it.'

Student Activity

Research and write the guide. Include free alternatives where possible. Format as a handout.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. Name one system software and one application software. 2. What is the difference between free and open-source software? 3. Why is antivirus important?’

Student Activity

Complete the exit ticket. Review software categories.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Copyright Basics - mathematical concepts
Science: Scientific applications of Copyright Basics - scientific applications
Language: Writing about Copyright Basics builds technical vocabulary and communication skills

Digital Citizen Reminder: Respect copyright - only use content you have permission to use.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Copyright Basics concepts through practical exercises
Advanced: Apply Copyright Basics concepts to solve complex problems independently
SEND: Practice Copyright Basics skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Copyright Basics activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Copyright Basics into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Copyright Basics. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.08 **Duration: 40 minutes**

Plagiarism

Learning Objectives

- Define plagiarism and understand why it's wrong
- Learn proper ways to credit original sources
- Practice paraphrasing and citing sources correctly

Engage (5-7 minutes)

Teacher Activity

Ask: ‘What software do you use every day? What would happen if all software disappeared?’ Discuss how software makes computers useful.

Student Activity

List software: browser, WhatsApp, games, calculator. Discuss dependency on software for daily tasks.

Explore (10-12 minutes)

Teacher Activity

Show categories of software: system OS, drivers, application browser, editor, utility antivirus, backup. Students categorize given examples.

Student Activity

Work in pairs. Sort 15 software names into correct categories. Discuss any disagreements.

Explain (10-12 minutes)

Teacher Activity

Define each category with examples: OS manages hardware, applications do specific tasks, utilities maintain the system. Show Linux as a free OS alternative.

Student Activity

Take notes. Understand the hierarchy: hardware ⟶ OS ⟶ applications. See Linux as an option.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Create a software guide for a new student. List 5 essential software for school work, explain what each does, and where to get it.’

Student Activity

Research and write the guide. Include free alternatives where possible. Format as a handout.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name one system software and one application software. 2. What is the difference between free and open-source software? 3. Why is antivirus important?'

Student Activity

Complete the exit ticket. Review software categories.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Plagiarism - mathematical concepts
Science: Scientific applications of Plagiarism - scientific applications
Language: Writing about Plagiarism builds technical vocabulary and communication skills

Digital Citizen Reminder: Always give credit to original authors - plagiarism is wrong.

Resources Needed:
- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Plagiarism concepts through practical exercises
Advanced: Apply Plagiarism concepts to solve complex problems independently
SEND: Practice Plagiarism skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Plagiarism activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Plagiarism into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Plagiarism. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.09 **Duration: 40 minutes**

Phishing Scams

Learning Objectives

- Understand what phishing is and how it works
- Identify common phishing tactics and warning signs
- Learn how to protect themselves from phishing attempts

Engage (5-7 minutes)

Teacher Activity

Ask: 'What software do you use every day? What would happen if all software disappeared?' Discuss how software makes computers useful.

Student Activity

List software: browser, WhatsApp, games, calculator. Discuss dependency on software for daily tasks.

Explore (10-12 minutes)

Teacher Activity

Show categories of software: system OS, drivers, application browser, editor, utility antivirus, backup. Students categorize given examples.

Student Activity

Work in pairs. Sort 15 software names into correct categories. Discuss any disagreements.

Explain (10-12 minutes)

Teacher Activity

Define each category with examples: OS manages hardware, applications do specific tasks, utilities maintain the system. Show Linux as a free OS alternative.

Student Activity

Take notes. Understand the hierarchy: hardware ⟶ OS ⟶ applications. See Linux as an option.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a software guide for a new student. List 5 essential software for school work, explain what each does, and where to get it.'

Student Activity

Research and write the guide. Include free alternatives where possible. Format as a handout.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name one system software and one application software. 2. What is the difference between free and open-source software? 3. Why is antivirus important?'

Student Activity

Complete the exit ticket. Review software categories.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Phishing Scams - mathematical concepts
Science: Scientific applications of Phishing Scams - scientific applications
Language: Writing about Phishing Scams builds technical vocabulary and communication skills

Digital Citizen Reminder: Be suspicious of messages asking for personal information - they may be scams.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Phishing Scams concepts through practical exercises
Advanced: Apply Phishing Scams concepts to solve complex problems independently
SEND: Practice Phishing Scams skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Phishing Scams activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Phishing Scams into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Phishing Scams. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.10 **Duration: 40 minutes**

Screen Time Management

Learning Objectives

- Understand the effects of excessive screen time on health
- Learn strategies for balanced technology use
- Create a personal screen time management plan

Engage (5-7 minutes)

Teacher Activity

Ask: 'What software do you use every day? What would happen if all software disappeared?' Discuss how software makes computers useful.

Student Activity

List software: browser, WhatsApp, games, calculator. Discuss dependency on software for daily tasks.

Explore (10-12 minutes)

Teacher Activity

Show categories of software: system OS, drivers, application browser, editor, utility antivirus, backup. Students categorize given examples.

Student Activity

Work in pairs. Sort 15 software names into correct categories. Discuss any disagreements.

Explain (10-12 minutes)

Teacher Activity

Define each category with examples: OS manages hardware, applications do specific tasks, utilities maintain the system. Show Linux as a free OS alternative.

Student Activity

Take notes. Understand the hierarchy: hardware → OS → applications. See Linux as an option.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a software guide for a new student. List 5 essential software for school work, explain what each does, and where to get it.'

Student Activity

Research and write the guide. Include free alternatives where possible. Format as a handout.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name one system software and one application software. 2. What is the difference between free and open-source software? 3. Why is antivirus important?'

Student Activity

Complete the exit ticket. Review software categories.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Screen Time Management - mathematical concepts
Science: Scientific applications of Screen Time Management - scientific applications
Language: Writing about Screen Time Management builds technical vocabulary and communication skills

Digital Citizen Reminder: Balance screen time with outdoor activities for healthy development.

Resources Needed:
- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Screen Time Management concepts through practical exercises
Advanced: Apply Screen Time Management concepts to solve complex problems independently
SEND: Practice Screen Time Management skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Screen Time Management activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Screen Time Management into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Screen Time Management. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.11 **Duration: 40 minutes**

Project - Safety Poster

Learning Objectives

- Apply digital citizenship knowledge to create an educational poster
- Design visual content that communicates safety messages effectively
- Present and explain their poster to classmates

Engage (5-7 minutes)

Teacher Activity

Ask: 'What software do you use every day? What would happen if all software disappeared?' Discuss how software makes computers useful.

Student Activity

List software: browser, WhatsApp, games, calculator. Discuss dependency on software for daily tasks.

Explore (10-12 minutes)

Teacher Activity

Show categories of software: system OS, drivers, application browser, editor, utility antivirus, backup. Students categorize given examples.

Student Activity

Work in pairs. Sort 15 software names into correct categories. Discuss any disagreements.

Explain (10-12 minutes)

Teacher Activity

Define each category with examples: OS manages hardware, applications do specific tasks, utilities maintain the system. Show Linux as a free OS alternative.

Student Activity

Take notes. Understand the hierarchy: hardware → OS → applications. See Linux as an option.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a software guide for a new student. List 5 essential software for school work, explain what each does, and where to get it.'

Student Activity

Research and write the guide. Include free alternatives where possible. Format as a handout.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name one system software and one application software. 2. What is the difference between free and open-source software? 3. Why is antivirus important?'

Student Activity

Complete the exit ticket. Review software categories.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Safety Poster - mathematical concepts
Science: Scientific applications of Project - Safety Poster - scientific applications
Language: Writing about Project - Safety Poster builds technical vocabulary and communication skills

Digital Citizen Reminder: Follow all online safety rules to protect yourself and your information.

Resources Needed:

- Chapter 6 reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Project - Safety Poster concepts through practical exercises
Advanced: Apply Project - Safety Poster concepts to solve complex problems independently
SEND: Practice Project - Safety Poster skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment
CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Safety Poster project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Safety Poster project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.12 **Duration: 40 minutes**

Chapter 6 Review

Learning Objectives

- Review and consolidate all digital citizenship concepts from Chapter 6
- Demonstrate understanding through a comprehensive review activity
- Create a personal action plan for responsible digital citizenship

Engage (5-7 minutes)

Teacher Activity

Ask: 'What software do you use every day? What would happen if all software disappeared?' Discuss how software makes computers useful.

Student Activity

List software: browser, WhatsApp, games, calculator. Discuss dependency on software for daily tasks.

Explore (10-12 minutes)

Teacher Activity

Show categories of software: system OS, drivers, application browser, editor, utility antivirus, backup. Students categorize given examples.

Student Activity

Work in pairs. Sort 15 software names into correct categories. Discuss any disagreements.

Explain (10-12 minutes)

Teacher Activity

Define each category with examples: OS manages hardware, applications do specific tasks, utilities maintain the system. Show Linux as a free OS alternative.

Student Activity

Take notes. Understand the hierarchy: hardware ⟶ OS ⟶ applications. See Linux as an option.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a software guide for a new student. List 5 essential software for school work, explain what each does, and where to get it.'

Student Activity

Research and write the guide. Include free alternatives where possible. Format as a handout.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name one system software and one application software. 2. What is the difference between free and open-source software? 3. Why is antivirus important?'

Student Activity

Complete the exit ticket. Review software categories.

Cross-Curricular Connections

Mathematics: Review exercises consolidate technology concepts
Science: Chapter summaries connect all emerging tech topics
Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 6 Review skills responsibly and ethically.

Resources Needed:
- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 6 Review support
On Level: Demonstrate Chapter 6 Review through practical exercises
Advanced: Apply Chapter 6 Review to solve complex problems independently
SEND: Practice Chapter 6 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 6 Review activity
CT Skill Assessed: Decomposition - breaking down Chapter 6 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 6 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson: